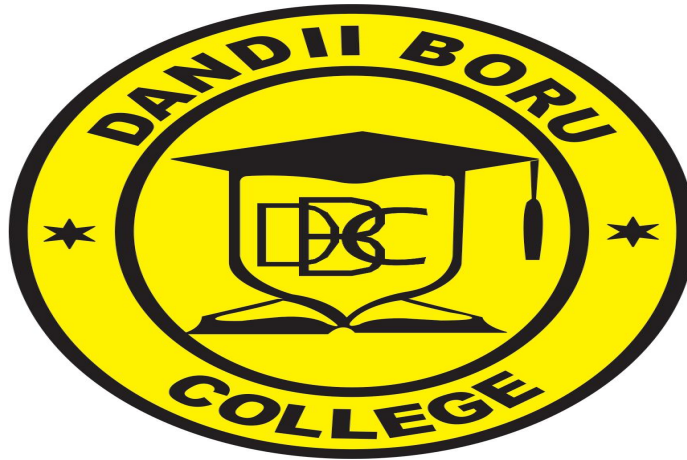




PHARMACOLOGY ONE
4 Cr.hr

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General pharmacology

INTRODUCTION

- **Pharmacology**
- can be defined as the study of **substances** that interact with living systems through chemical processes,
 - ✓ especially by binding to **regulatory molecules** and **activating** or **inhibiting** normal body processes.
 - ✓ These substances may be chemicals administered to achieve a **beneficial therapeutic effect** on some process within the patient
 - ✓ is the study of drugs and their interaction with living systems.
 - ✓ It studies what our body does to the drug and what the drug does to our body

INTRODUCTION

- **Drug** - a substance (chemical agent) that affects a biological system in a potentially useful way.
 - ✓ Drugs are used in the prevention, diagnosis, treatment or cure of disease in man or other animals.
- **Poison** - Any chemical agent that produces a harmful effect
- **Toxin** - a poison of biological origin
- A **medicine** is a chemical preparation, which usually but not necessarily contains one or more drugs, administered with the intention of producing a therapeutic effect.

History of pharmacology

- 17th century: reliance on **observation** and **experimentation** began to replace theorizing in medicine
- Late 18th and early 19th centuries, François Magendie, and later his student Claude
- Bernard, began to develop the methods of experimental physiology and pharmacology.
 - ✓ Clinical trials: 1950's
 - ✓ Drug receptor
 - ✓ Pharmacogenetics

History of pharmacology

- The extension of scientific principles into everyday therapeutics is still going on, although the medication-consuming public is still exposed to vast amounts of **inaccurate, incomplete, or unscientific** information regarding the pharmacologic effects of chemicals.
- This has resulted in the **irrational use of innumerable expensive, ineffective, and sometimes harmful remedies and the growth of a huge “alternative health care” industry**

Branch of pharmacology

❖ Two Primary Branches of Pharmacology

- I. Pharmacodynamics:-
 - ✓ The study of the relationship between concentrations of drug and the biologic effects (physiological or biochemical) with time.
 - ✓ “what drugs do to the body”.
- II. Pharmacokinetics:-
 - ✓ The study of the absorption, distribution, metabolism and excretion of drugs from the body.
 - ✓ “what the body does to drugs”.

Branch of pharmacology

■ Toxicology :

- ✓ the science of poisons
- ✓ study of the undesirable effects of chemical agents on living systems.
- ✓ Poisons are substances that cause harmful, dangerous or fatal symptoms in living things
- ✓ Many drugs in larger doses may act as poisons
- ✓ study of the [adverse effects](#), molecular targets, and characterization of drugs or any chemical substance in **excess**

Pharmacogenetics

- ✓ study of genetic variation which cause different response to drugs
- ✓ is clinical testing of genetic variation that gives rise to differing response to drugs.

- Pharmacogenomics

- ✓ is the application of genomic technologies to drug discovery and further characterization of older drugs.
- ✓ investigation in variation in DNA and RNA characteristics related to drug response

Pharmacoepidemiology

- ✓ is the study of the effects of drugs in large numbers of people.
- ✓ It includes trial studies on healthy and diseased individuals and collecting the opinion of physicians prescribing a drug.

Pharmacovigilance

- ✓ science of collecting, monitoring and evaluating information of the adverse effects of medications, biological products, herbalism and other medicines

Pharmacoeconomics

It is the study of economic factors regarding the cost of drug therapies, including their impact on healthcare system and society.

Drug:

- The word [drug](#) comes from a French word '*Droque*' meaning a dry herb.
- It can be defined as: “Substance or material that is used or intended to be used to modify or explore
 - ✓ physiological processes or
 - ✓ pathological states, for the benefit of the recipient.”

Source of drugs

- **Minerals:** Liquid paraffin, magnesium sulfate, magnesium trisilicate, kaolin, etc.
- **Animals:** Insulin, thyroid extract, heparin and antitoxin sera, etc.
- **Plants:** Morphine, digoxin, atropine, castor oil, etc.
- **Synthetic source:** Aspirin, sulphonamides, paracetamol, zidovudine, etc.
- **Micro organisms:** Penicillin, streptomycin and many other antibiotics.
- **Genetic engineering:** Human insulin, human growth hormone etc

Drug nomenclature

- A drug generally has three categories of names:
- (a) **Chemical name:**
- A description of that drug using the nomenclature of chemistry e.g. 1-(Isopropylamino)-3-(1-naphthyloxy)
- Chemical names are **too complex** to be used in prescription
- A code name, e.g. RO 15-1788 (later named flumazenil) may be assigned by the manufacturer for convenience and simplicity before an approved name is coined.

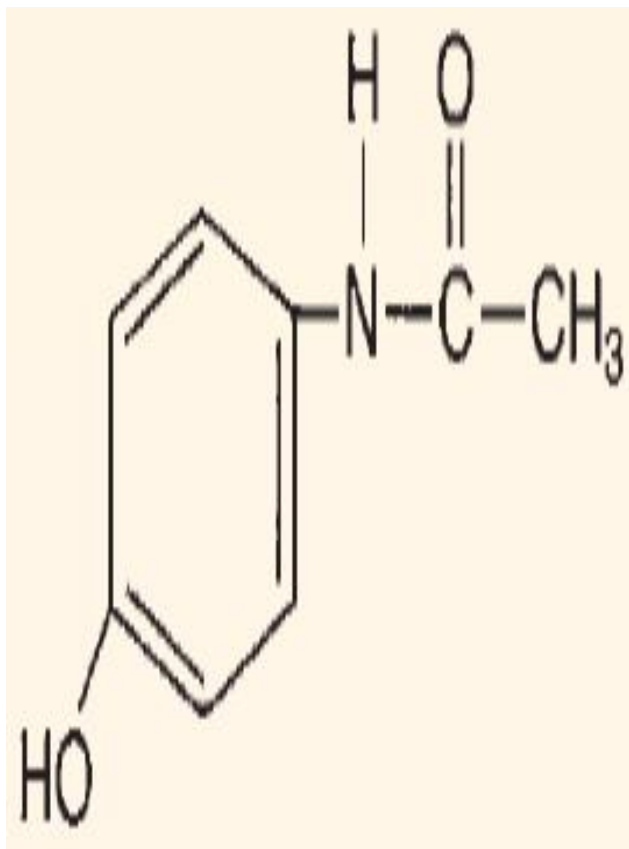
Drug nomenclature

- **(b) Non-proprietary(generic) name**
- It is the name accepted by a **competent scientific body/ authority**
 - ✓ Assigned by WHO
- Internationally accepted name
- Until the drug is included in a pharmacopoeia, the nonproprietary name may also be called the **approved name**.
- After its appearance in the official publication, it becomes the **official name**.

Drug nomenclature

- **(c) Proprietary (Brand) name**
 - ✓ It is the name assigned by the **manufacturer(s)** and is his property or trade mark.
 - ✓ One drug may have **multiple proprietary names**,
 - ✓ Brand names are designed to be
 - ✓ **catchy , short, easy to remember and often suggestive**
 - ✓ In addition, combined formulations have their own **multiple brand** names.

Drug nomenclature



Type of drug name	Examples
Chemical name	N-acetyl-para-aminophenol
Generic name	Paracetamol
Trade name	Panadol, Tylenol, Dapa, Valadol, Valorin, Helenol, etc

New Drug Development

- ✓ Is an expensive and lengthy process
- ✓ It comprises of two steps

a) Preclinical development

- ✓ the aim is to explore the biological effects (**efficacy** and **safety**) of a new substance before it is administrated to patients
- ✓ varying drug **doses** are tested on animals and/or in vitro systems

New Drug Development Cont...

b) Clinical development

- ✓ Includes testing on **human beings** to determine tolerable **dose**, **duration** of action, comparing it with standard treatment.
- ✓ Then **licensing for market** and **post marketing surveillance** will be done
- The clinical development of new drugs usually takes place in steps or phases
 - ✓ clinical pharmacology (*phase I*),
 - ✓ clinical investigation (*phase II*),
 - ✓ clinical trials (*phase III*),
 - ✓ postmarketing studies (*phase IV*)

Phase I

- ✓ administered to humans for the first time,
- ✓ conducted in healthy men between 18 and 45 years of age
- ✓ For antineoplastic agents, it is not appropriate to use healthy men
- purpose **:is to establish the dose level at which signs of toxicity first appear.**
- administering a **single dose** of the test drug and **closely observing**
- If no adverse reactions occur, the dose is increased.
- confined to a group of 20 to 80 subjects.
- If no untoward effects result from single doses, short-term **multiple-dose** studies are initiated.

Phase II

- If the results of phase I studies show that it is reasonably safe to continue, the new drug is administered to **patients** for the first time.
- Efforts are concentrated on evaluating **efficacy** and **on establishing an optimal dose range**.
- Therefore, **dose–response studies** are a critical part of phase II studies.
- The number of subjects between 80 and 100.

Phase III

- large numbers of subjects can be exposed to the drug.
- subjects may range from several hundred to several thousand, depending on the drug.
- The purpose: is to verify the **efficacy** of the drug and to detect **effects** that may not have surfaced in the phase I and II trials, during which exposure to the drug was limited.
- Under this procedure, drugs can be approved on the basis of phase II studies conducted in a limited number of patients.

Phase IV

- Controlled and uncontrolled studies often are conducted after a drug is approved and marketed.
- Such studies are intended to broaden the experience with the drug and compare it with other drugs.

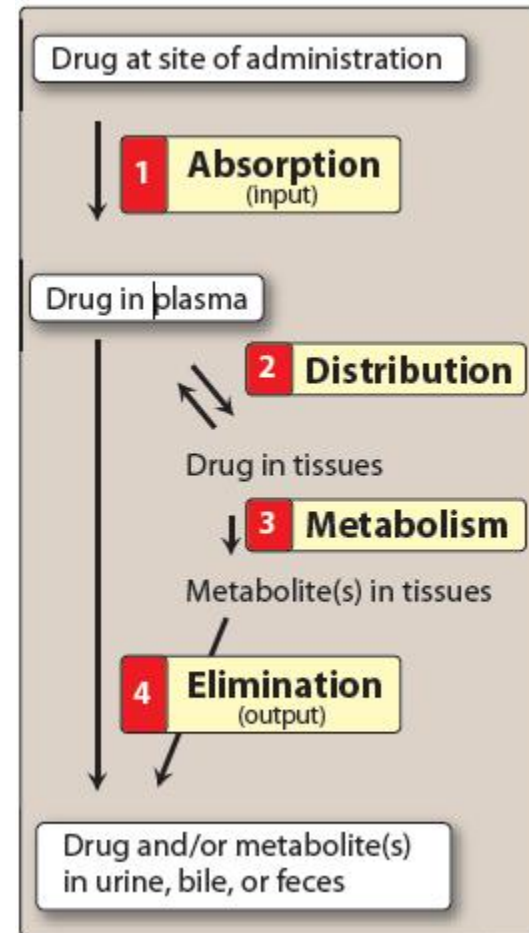
Summery of Phases of Clinical Investigation

Phase	Purpose
I	Establish safety
II	Establish efficacy and dose
III	Verify efficacy and detect adverse affects
IV	Obtain additional data following approval

Pharmacokinetics

❖ Pharmacokinetics refers to:-

- what the body does to a drug
- how a drug reaches its target in the body and how it is affected on that journey,
- how is the drug absorbed, distributed, metabolized and excreted in the body
 - ✓ the speed of onset of drug action,
 - ✓ the intensity of the drug's effect,
 - ✓ and the duration of drug action



Absorption

- **Absorption:** is the transfer of a drug from its site of administration to the bloodstream
- The rate and efficiency of absorption depend on
 - ✓ environmental factors
 - ✓ the drug's chemical characteristics factors
 - ✓ route of administration factors
- Except when given i.v., the drug has to cross biological membranes, For IV delivery, absorption is complete.
 - ✓ Drug delivery by other routes may result in only partial absorption and, thus, lower bioavailability.

biological membrane

- **Structure of biological membrane:**
- The outer surface of the cell covered by a very thin structure known as **plasma membrane**.
- It is composed of lipid and protein molecules.
- The plasma membrane is a **semipermeable** membrane allowing certain chemical substances to pass freely
 - ✓ it allows water, glucose, etc.
 - ✓ it won't allow sucrose until it is converted into glucose and fructose

biological membrane

- The membrane proteins have many functions like
 - ✓ (a) contributing structure to the membrane
 - ✓ (b) acting as enzyme
 - ✓ (c) acting as carrier for transport of substances
 - ✓ (d) acting as receptors.

Route of administration

ROUTE OF ADMINISTRATION	ABSORPTION PATTERN	ADVANTAGES	DISADVANTAGES
Oral	● Variable; affected by many factors	● Safest and most common, convenient, and economical route of administration	● Limited absorption of some drugs ● Food may affect absorption ● Patient compliance is necessary ● Drugs may be metabolized before systemic absorption
Intravenous	● Absorption not required	● Can have immediate effects ● Ideal if dosed in large volumes ● Suitable for irritating substances and complex mixtures ● Valuable in emergency situations ● Dosage titration permissible ● Ideal for high-molecular-weight proteins and peptide drugs	● Unsuitable for oily or poorly absorbed substances ● Bolus injection may result in adverse effects ● Most substances must be slowly injected ● Strict aseptic techniques needed
Subcutaneous	● Depends on drug diluents: Aqueous solution: prompt Depot preparations: slow and sustained	● Suitable for slow-release drugs ● Ideal for some poorly soluble suspensions	● Pain or necrosis if drug is irritating ● Unsuitable for drugs administered in large volumes
Intramuscular	● Depends on drug diluents: Aqueous solution: prompt Depot preparations: slow and sustained	● Suitable if drug volume is moderate ● Suitable for oily vehicles and certain irritating substances ● Preferable to intravenous if patient must self administer	● Affects certain lab tests (creatinine kinase) ● Can be painful ● Can cause intramuscular hemorrhage (precluded during anticoagulation therapy)
Transdermal (patch)	● Slow and sustained	● Bypasses the first-pass effect ● Convenient and painless ● Ideal for drugs that are lipophilic, thus requiring prolonged administration ● Ideal for drugs that are quickly eliminated from the body	● Some patients are allergic to patches, which can cause irritation ● Drug must be highly lipophilic ● May cause delayed delivery of drug to pharmacological site of action ● Limited to drugs that can be taken in small daily doses

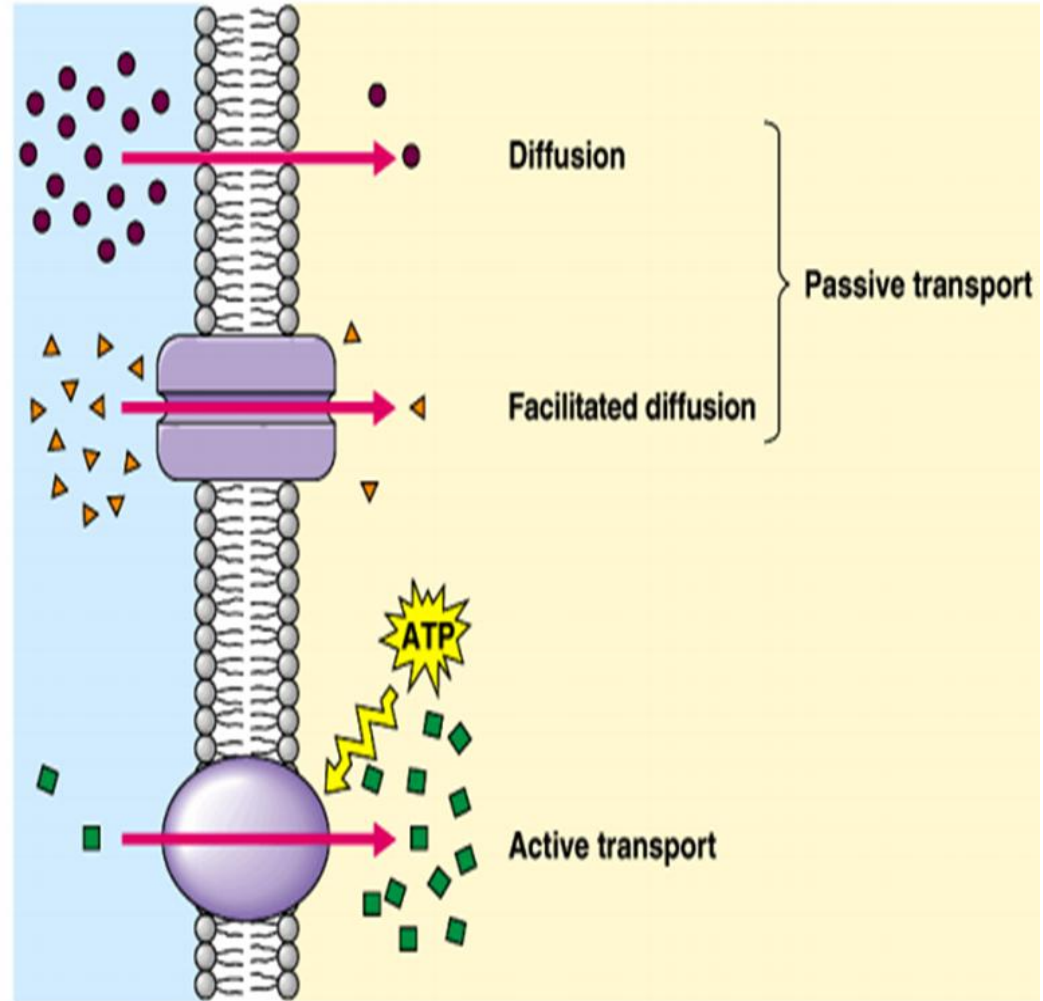
Route of administration

Rectal	● Erratic and variable	● Partially bypasses first-pass effect ● Bypasses destruction by stomach acid ● Ideal if drug causes vomiting ● Ideal in patients who are vomiting, or comatose	● Drugs may irritate the rectal mucosa ● Not a well-accepted route.
Inhalation	● Systemic absorption may occur. This is not always desirable	● Absorption is rapid; can have immediate effects ● Ideal for gases ● Effective for patients with respiratory problems ● Dose can be titrated ● Localized effect to target lungs: lower doses used compared to that with oral or parental administration ● Fewer systemic side effects	● Most addictive route (drug can enter the brain quickly) ● Patient may have difficulty regulating dose ● Some patients may have difficulty using inhalers
Sublingual	● Depends on the drug: Few drugs (for example, <i>nitroglycerin</i>) have rapid, direct systemic absorption Most drugs erratically or incompletely absorbed	● Bypasses first-pass effect ● Bypasses destruction by stomach acid ● Drug stability maintained because the pH of saliva relatively neutral ● May cause immediate pharmacological effects	● Limited to certain types of drugs ● Limited to drugs that can be taken in small doses ● May lose part of the drug dose if swallowed

Mechanisms of absorption of drugs from the GI tract

❖ drugs may be absorbed from the GI tract by:-

- Passive transfer
 - ✓ simple diffusion
 - ✓ filtration
- Specialized transport
 - ✓ i) Facilitated diffusion
 - ✓ ii) Active transport
 - ✓ iii) Endocytosis



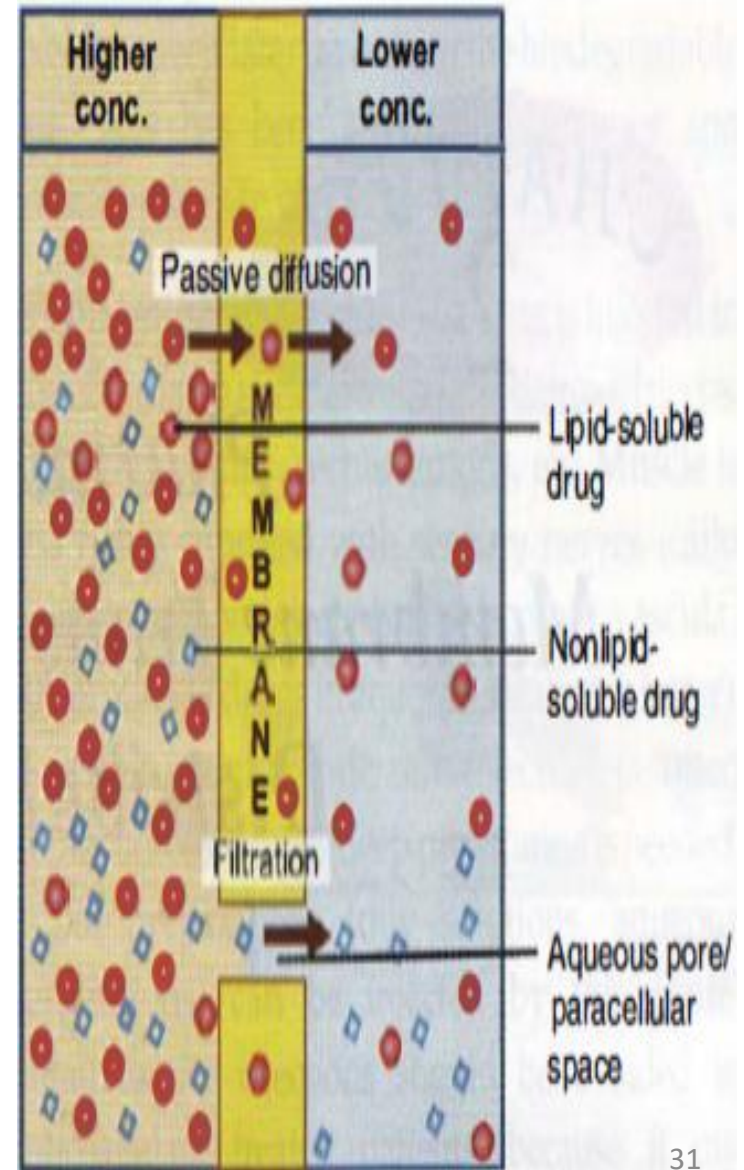
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1. Simple diffusion:

- The drug diffuses across the membrane in the direction of its concentration gradient
 - ✓ from a region of high concentration to one of lower concentration.
- The driving force is the concentration gradient
- Does not involve a carrier,
- Can not be saturated
- no energy source is required,
- also called non-ionic diffusion, e.g. highly lipid soluble drugs

Simple diffusion cont...

- The vast majority of drugs gain access to the body by this mechanism.
- lipid-soluble drugs readily move across most biologic membranes due to their solubility in the membrane lipid bilayers

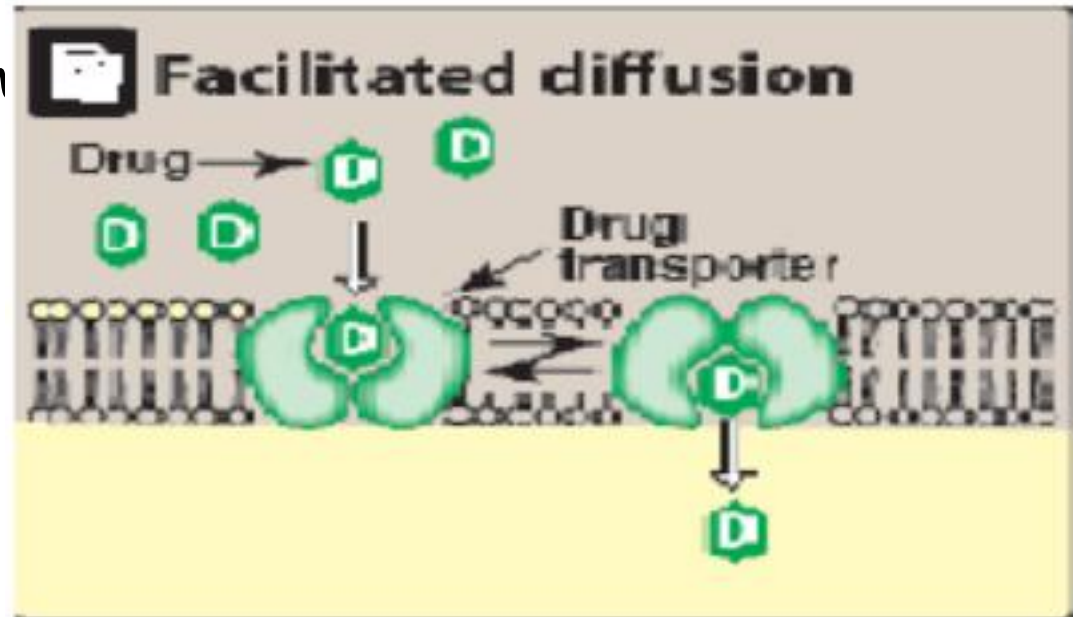


- Filtration

- ✓ passage of drugs through **aqueous pores** in the membrane or through Para cellular spaces.
- ✓ **Lipid-insoluble drugs** ,E.g. urea, water and sugar pass via pores
 - It must be low molecular weight and low molecular size
 - ✓ less than pore size.
- Majority of cells have very small pores and drugs with MW > 100 or 200 are not able to penetrate.
- The driving force is hydrostatic (hydrodynamic) pressure gradient or osmotic pressure gradient

- Specialized transport
- a component of the membrane called as the carrier
 - ✓ that binds reversibly or non-covalently with the solute molecules to be transported.
- The carrier-solute complex traverses across the membrane to the other side
 - ✓ where it discharges the solute molecule. Eg. Water soluble monosaccharide, aminoacids, vitamins.
- ❖ **Two type of carrier transport**
 - ✓ Facilitated diffusion
 - ✓ Active transport

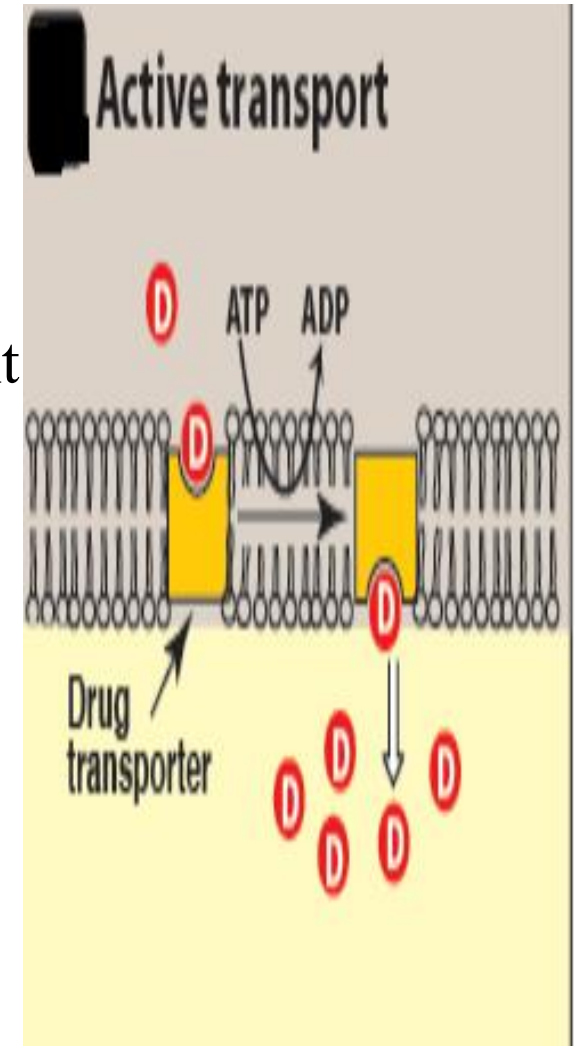
- *Facilitated diffusion:*
- carrier proteins that facilitate the passage of large molecules.
- from high concentration to of low concentration.
- It does not require energy,
- can be saturated, and may be inhibited by compounds that com



Active transport:

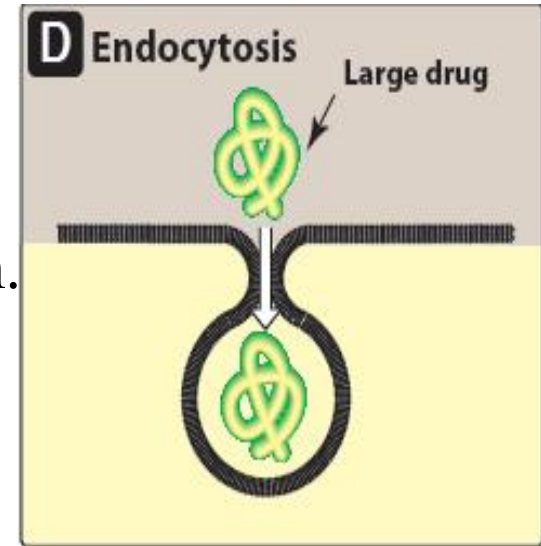
- This mode of drug entry also involves **specific carrier proteins** that span the membrane.
- Energy-dependent active transport
- moving drugs against a concentration gradient
- shows saturation kinetics for the carrier
- Active transport systems are selective
- ✓ competitively inhibited by:

other cotransported substances.



- **Endocytosis:** transport drugs, exceptionally large size
- It is minor transport mechanism engulfing extracellular materials with in a segment of the cell membrane to form a saccule which is then pinched off intracellularly.

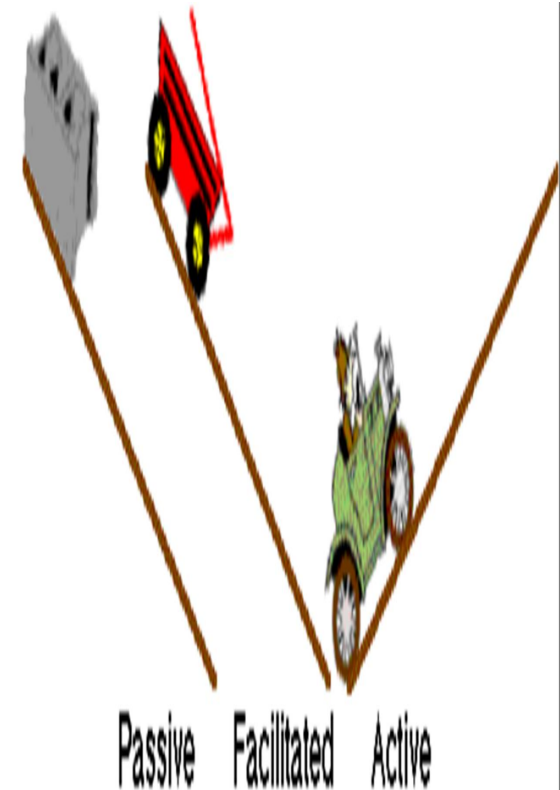
Eg. Fat and starch, oil soluble vitamins and insulin.



- Endocytosis includes **two types**
- ✓ **Phagocytosis (cell eating):** adsorptive uptake of solid particulates.
- ✓ **Pinocytosis:** uptakes of fluid solute. Eg. Oral polio vaccine

Summary of drug transport mechanisms

Cxs	Simple diffusion	Facilitated	Active transport
Incidence	commonest	less common	least common
Process	slow	quick	very quick
Movement	along the conc. gradient	along the conc. gradient	against conc. gradient
Carrier	not needed	needed	needed
Energy	not required	not required	required

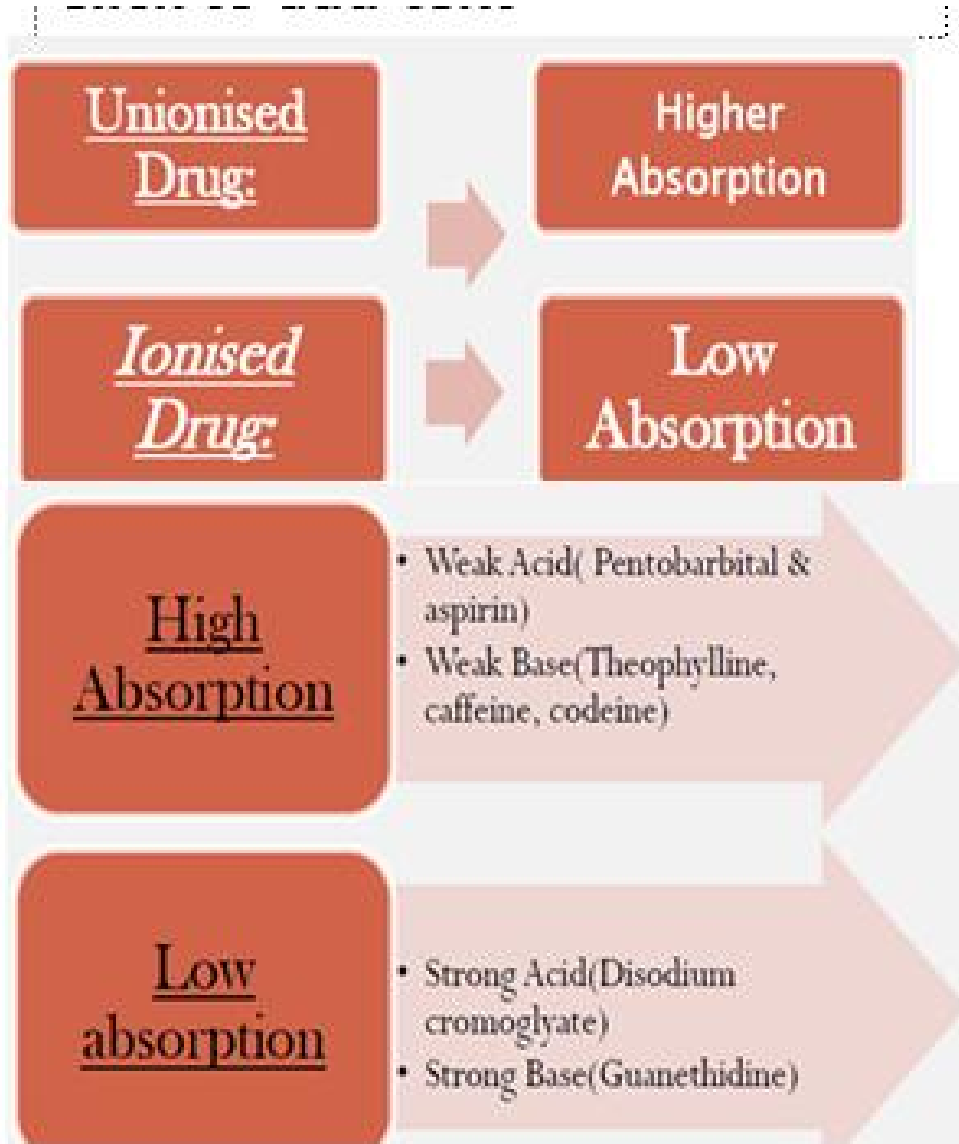


Factors affecting drug absorption

Effect of pH on drug absorption:

- Most of the drugs are weak acids or weak bases,
- **Unionized** component is predominantly **lipid soluble** and is **absorbed rapidly**
- **ionized** is often **water soluble component** which is **absorbed poorly**, the mucosal lining of the G.I.T is impermeable to the ionized form
- **Acidic drugs**: rapidly absorbed from the stomach e.g. salicylates
- **Basic drugs**: absorbed in alkaline environment i.e. small intestine e.g. pethidine and ephedrine.

Effect of pH on drug absorption:



Effect of food

- Food can affect absorption by Altering,
 - ✓ ∂ Stomach pH
 - ✓ ∂ Affect Gastric emptying
 - ✓ ∂ Forms complex with the drug.

Gastric transit time

- is the passage from stomach to small intestine.
- Rapid absorption, when the drug is given on empty stomach.
- certain irritant drugs like salicylates and iron administered after food
- sometimes the presence of food in the G.I tract aids the absorption of certain drugs e.g. griseofulvin, propranolol and riboflavin.

Factor	Increased gastric emptying rate	Decreased gastric emptying rate
Physiological	Liquids, gastric distention	Solids, acids, fat
Pathological	Duodenal ulcers, gastroenterostomy, chronic pancreatitis	Acute abdominal trauma and pain, labor of child birth, gastric juices, intestinal obstruction, pneumonia, diabetes mellitus
Pharmacological	Reserpine, anticholinesterases, guanethidine, cholinergic agents	Anticholinergic drugs, ganglionic blocking drugs, narcotic analgesics

GI motility and Presence of other agents:

- GI motility tends to move the drug through the alimentary canal so that the drug may not stay at the absorption site.
- e.g. cathartic drug increase gastric motility & decrease drug absorption of another drug.
- **Presence of other agents:**
- Vitamin C enhances the absorption of iron from the G.I.T.
- Calcium present in milk and in antacids forms insoluble complexes with the tetracycline, reduces their absorption

Blood flow to the absorption site:

- Because blood flow to the intestine is **much greater** than the flow to the stomach,
 - ✓ absorption from the intestine is favored over that from the stomach.
- Note: Shock severely reduces blood flow to cutaneous tissues, thereby minimizing the absorption from SC administration.

■ Age:

- In children & Infants
 - ✓ Gastric pH is high,
 - ✓ membrane & BBB permeability is high,
 - ✓ protein binding is less therefore it imparts drug absorption.
- while in Elderly patient there is altered gastric emptying,
 - ✓ decrease intestinal surface area,
 - ✓ decrease gastric blood flow
 - ✓ higher incidence of achlorhydria so it imparts drug absorption.

Diseases condition

- i) Gastric disease
- ii) Intestinal diseases
- iii) Cardio-vascular diseases
- Several changes associated with congestive cardiac failure influence the bio-availability of the drug
 - ✓ edema of the intestine,
 - ✓ decreases blood flow to GIT, etc.
- iv) Hepatic diseases
 - ✓ hepatic cirrhosis influence bio-availability

Interactions

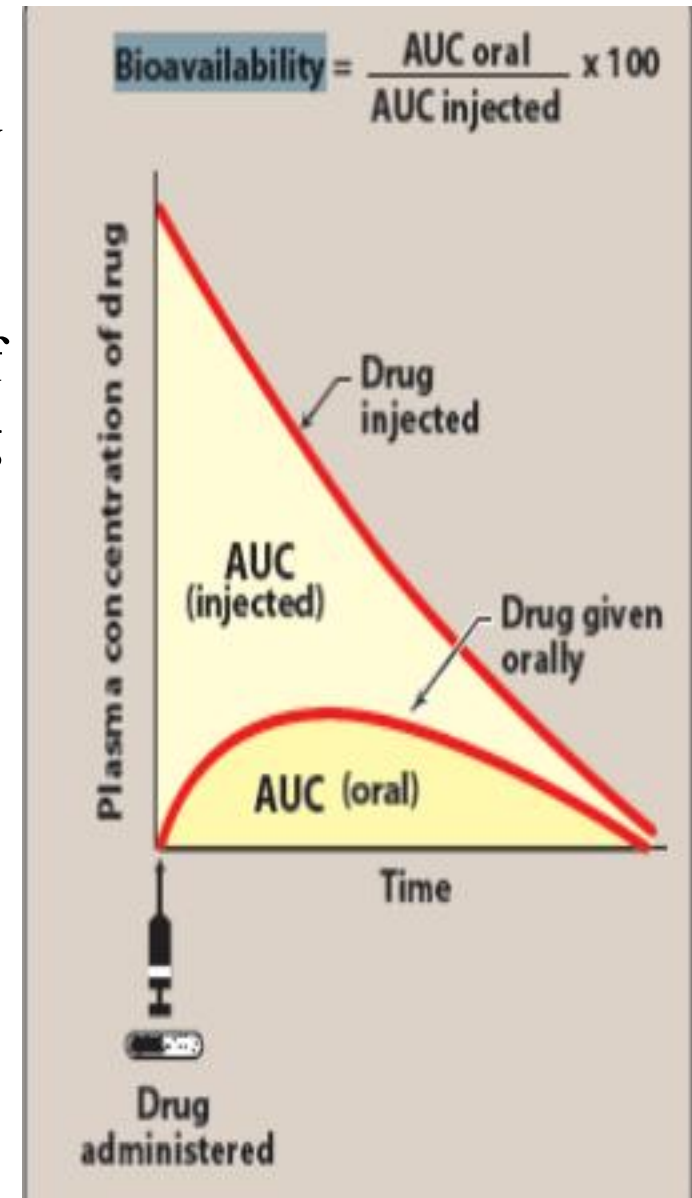
❖ i) Food-Drug Interactions

- ii) Drug-Drug Interactions
- iii) Drug-GI contents interactions
- Interaction with mucin, enzymes and bile salts influence drug absorption.
- E.g. Bile salts
 - ✓ Increase absorption of –Vitamins (Solution)
 - ✓ Decrease absorption of –Neomycin and Kanamycin (Insoluble complex)

Bioavailability:

- It is the rate and amount of drug that is absorbed from a given dosage form and reaches the systemic circulation following
 —→ non-vascular administration
- When the drug is given IV, the bioavailability is 100%.
- The route of administration largely determines the latent period between administration and onset of action.

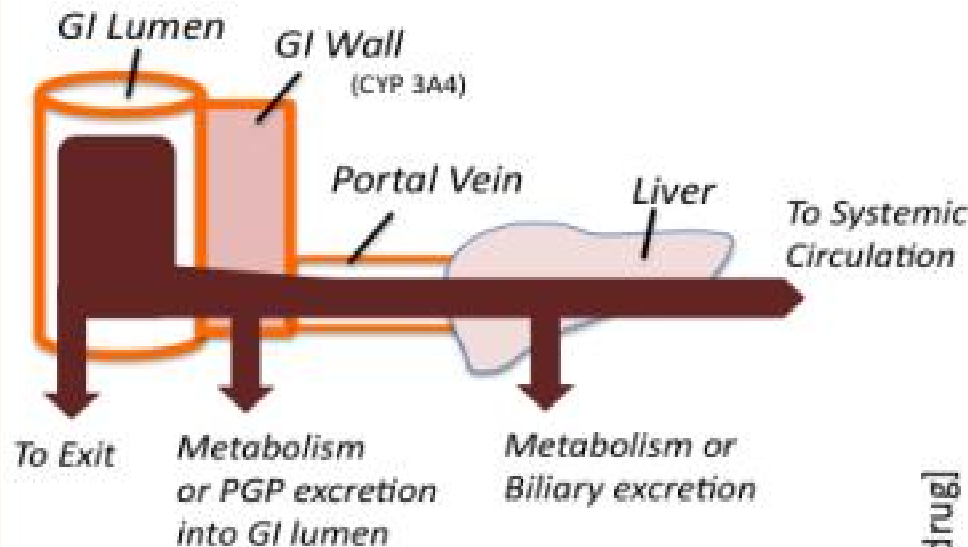
- **Determination of bioavailability:**
- Bioavailability is determined by comparing plasma levels of a drug
 - ✓ after a particular route of administration with plasma drug levels
- Bioavailability of a drug administered orally is
 - ✓ the ratio of the **area** calculated for oral administration compared with the **area** calculated for IV injection if doses are equivalent



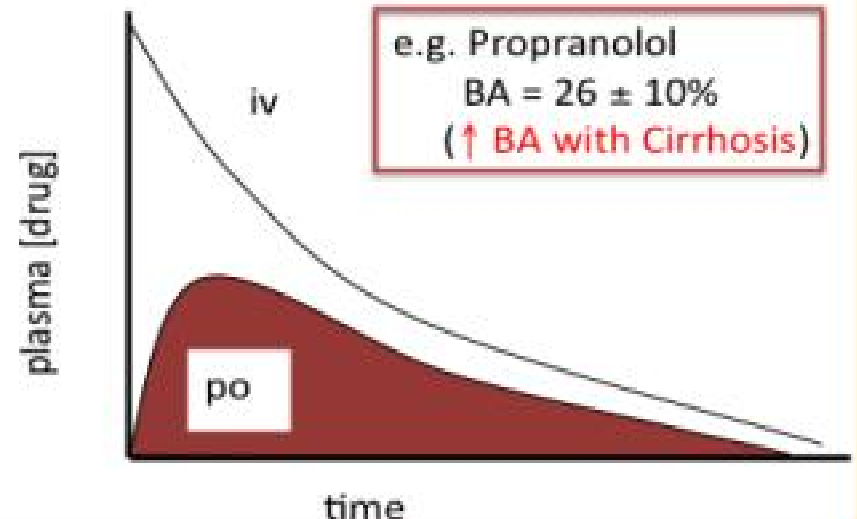
Summary: bioavailability

Bioavailability

- the fraction absorbed into the systemic circulation is the drug's bioavailability

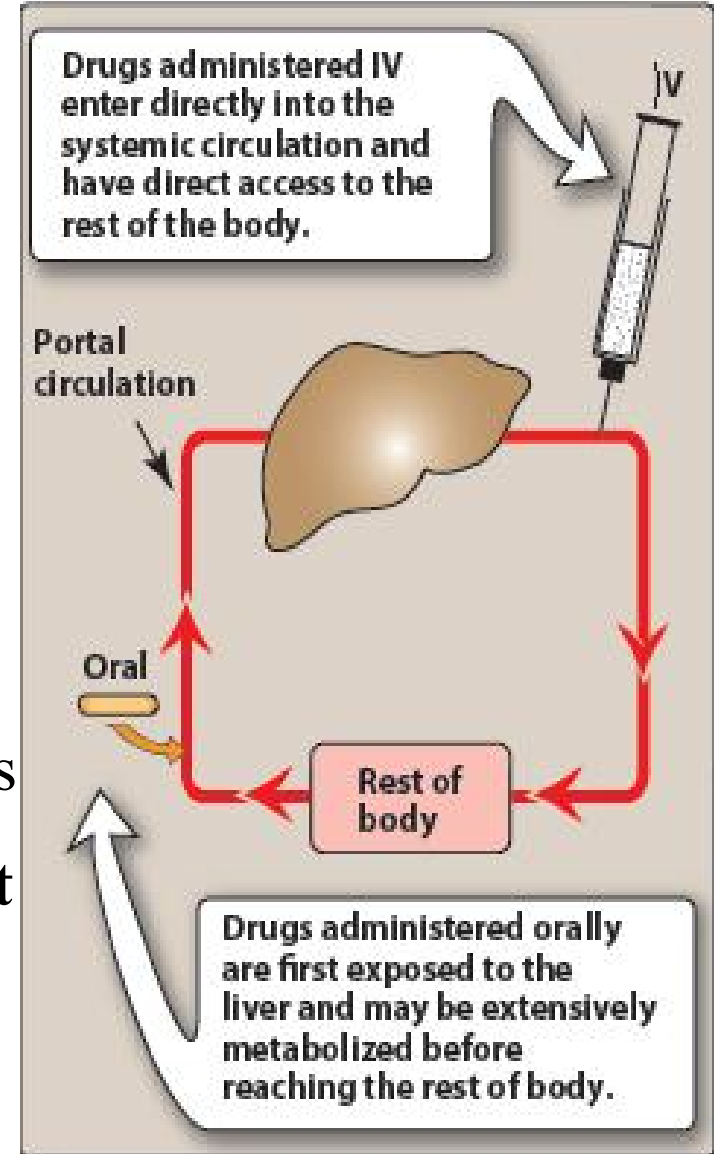


$$BA = \frac{AUC_{po}}{AUC_{iv}} \times 100$$



Factors that influence bioavailability

- **a. First-pass hepatic metabolism**
- If the drug is **rapidly metabolized** in the liver or gut wall during this initial passage → unchanged drug that gains **access to the systemic circulation** is **decreased**.
- Drugs that exhibit high first-pass metabolism should be given in **sufficient quantities**



- **b. Solubility of the drug:**
- Very hydrophilic drugs are **poorly absorbed**
 - ✓ because of their **inability to cross the lipid-rich** cell membranes.
- drugs that are extremely hydrophobic are also **poorly absorbed**,
 - ✓ because they are totally **insoluble in aqueous body fluids** and, therefore, cannot gain access to the surface of cells.
- For a drug to be readily absorbed, it must be:-
 - ✓ largely hydrophobic,
 - ✓ yet have some solubility in aqueous solutions.

- **c. Chemical instability:**

- Some drugs, such as *penicillin G*, are unstable in the pH of the gastric contents.

- **d. Nature of the drug formulation:**

- Drug absorption may be altered by factors **unrelated to the chemistry of the drug.**

- particle size, salt form, crystal polymorphism, enteric coatings, and the presence of excipients

—————→ can influence the ease of **dissolution** and, therefore, alter the rate of absorption

Distribution of drugs

- Once a drug has gained access to the blood stream,
 - ✓ it gets distributed to other tissues that initially had no drug,
- The extent of distribution of a drug depends on its
 - ✓ lipid solubility,
 - ✓ ionization at physiological pH,
 - ✓ extent of binding to plasma and tissue proteins,
 - ✓ differences in regional blood flow.

Distribution of drugs

- Penetration of a drug to the sites of action through the walls of blood vessels from the administered site after absorption
- Drugs distribute through various body fluid compartments such as
 - ✓ (a) plasma
 - ✓ (b) interstitial fluid compartment
 - ✓ (c) trans-cellular compartment

Distribution of drugs

High V_d

- High tissue protein binding
- Relatively high lipid solubility
- Distributed intracellularly
- Low plasma protein binding
- Difficult to efficiently removed by haemodialysis

Low V_d

- Confined to ECF
- Polar compounds
- High MW
- High plasma protein binding
- Do not cross BBB or placental barriers

Factors influencing drug distribution

- Once a drug has entered the blood compartment, the *rate* at which it penetrates tissues and other body fluids depends on several factors.
- These include
 - ✓ capillary permeability,
 - ✓ blood flow–tissue mass ratio (i.e., perfusion rate),
 - ✓ extent of plasma protein and specific organ binding,
 - ✓ regional differences in pH,
 - ✓ transport mechanisms available, and
 - ✓ the permeability characteristics of specific tissue membranes

Factors influencing drug distribution

❖ Tissue perfusion

- ✓  blood flow to tissues  distribution to interstitial fluids
- ✓ Drugs distribute more rapidly to lung, brain, liver and kidney than skeletal muscles & fat

❖ Physiochemical properties of the drug

- ✓ Molecular weight
- ✓ Pka of the drug and Regional differences in pH
- ✓ Lipid solubility

❖ Capillary Permeability

- ✓ BBB, Placental barrier

❖ Plasma protein binding and tissue protein binding

Factors influencing drug distribution

- **Plasma protein binding**
- Drug in systemic circulation exist as bound and unbound form in equilibrium
- $D + P \rightarrow [DP] \rightarrow D + P$
- Albumin
 - ✓ Major carrier for acidic drugs
- α 1-Acid Glycoprotein
 - ✓ Involved in the binding of basic drugs e.g. imipramine
- Others: lipoprotein, globulin

Factors influencing drug distribution

- **Tissues Binding**
- Drug concentrations will vary from organ to organ
 - ✓ Kidney: Gentamycin
 - ✓ liver: Chloroquine
 - ✓ Bone-: TTC, lead & other heavy metals
 - ✓ Fat: Drugs with extremely high lipid–water partition coefficients have a tendency to accumulate in body fat e.g. thiopental

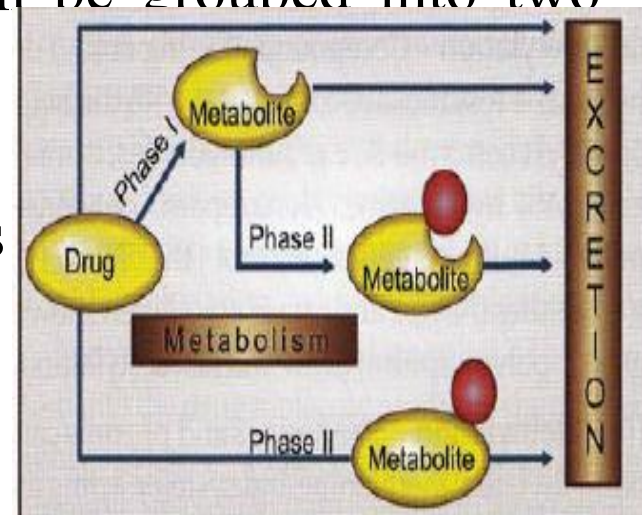
Factors influencing drug distribution

❖ Blood Brain Barrier

- The continuous tight junctions of brain capillary endothelial cells and pericapillary glial cells constitute the BBB
- Prevents the passage of drugs into the CNS
 - ✓ Only lipid soluble drugs or carrier mediated transport can cross BBB
 - ✓ Hydrophilic drugs can't cross BBB
 - ✓ Inflammation as in **meningitis** increase permeability to **ionized, lipid insoluble** drugs

Metabolism of Drugs

- The kidney cannot efficiently eliminate **lipophilic** drugs
- Therefore, lipid-soluble agents must first be metabolized into more polar (hydrophilic) substances in the liver
 - ✓ therefore more easily excreted.
- Drug metabolism reactions are carried out by enzyme systems.
- The enzyme systems for this purpose can be grouped into two categories:
 - ✓ phase I: oxidative or reductive enzymes
 - ✓ phase II: conjugative enzymes.

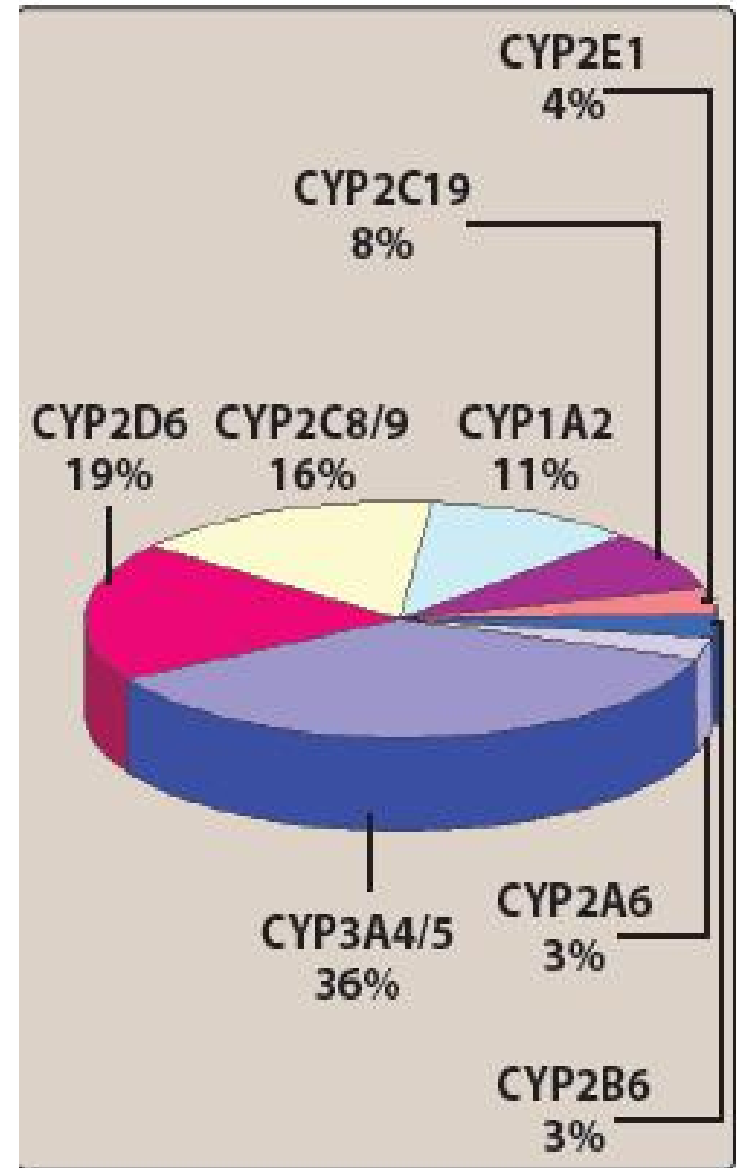


Phase I:

- Phase I reactions convert lipophilic molecules into more polar molecules by:-
 - ✓ introducing or
 - ✓ unmasking a polar functional group, such as -OH or -NH_2 .
- may increase, decrease, or leave unaltered the drug's pharmacologic activity.
- **Type of Phase - I reactions**
 - a) **Oxidation:** introduction of an oxygen/the removal of hydrogen
 - b) **Reduction:** The reduction reaction will take place
 - c) **Hydrolysis:** It means splitting of drug molecule after adding

Phase I:

- **Cytochrome P450 Enzymes**
- The cytochrome P450 (CYP450) enzyme is the primary phase I enzyme system
- These enzymes also are responsible for all or part of
 - ✓ metabolism
 - ✓ synthesis endogenous compounds,
 - E.g steroid hormones and prostaglandins.



Phase I:

- **Inducers:** The CYP450–dependent enzymes are an important target for **pharmacokinetic drug interactions**.
- interaction is the induction of selected CYP isozymes.
- Xenobiotics may induce the activity of these enzymes by inducing the expression of the genes encoding

Isozyme: CYP3A4/5

COMMON SUBSTRATES	INDUCERS
Carbamazepine Cyclosporine Erythromycin Nifedipine Verapamil	Carbamazepine Dexamethasone Phenobarbital Phenytoin Rifampin

Isozyme: CYP2C9/10

COMMON SUBSTRATES	INDUCERS
Warfarin Phenytoin Ibuprofen Tolbutamide	Phenobarbital Rifampin

Phase I

- **Inhibitors:** Inhibition of CYP isozyme activity is an important source of drug interactions
- The most common form of inhibition is through competition for the **same isozyme**.
- Numerous drugs have been shown to inhibit one or more of the CYP-dependent biotransformation pathways of *warfarin*.

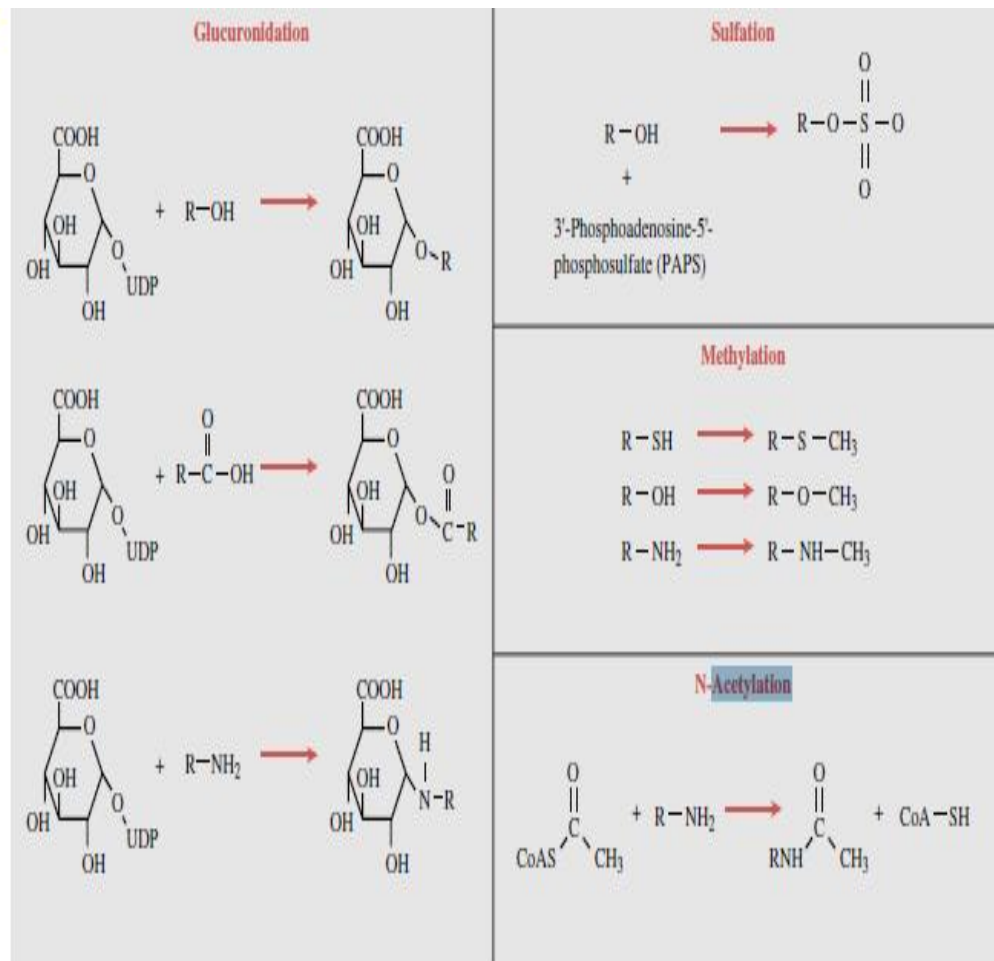
CYP Isoform	Examples of Inhibitors
CYP1A1	Thought to be same as CYP1A2
CYP1A2	Amiodarone, fluoroquinolone antibiotics, fluvoxamine
CYP2A6	Tranlycypromine, methoxsalen
CYP2B6	Efavirenz, nelfinavir, ritonavir
CYP2C8	Probably similar to CYP2C9
CYP2C9	Amiodarone, fluconazole, fluvastatin, lovastatin, zafirlukast
CYP2C19	Cimetidine, ketoconazole, omeprazole, ticlopidine ^a
CYP2D6	Amiodarone, cimetidine, fluoxetine, paroxetine, quinidine
CYP2E1	Disulfiram ^a
CYP3A4	HIV antivirals (e.g., Ritonavir), amiodarone, cimetidine, diltiazem, erythromycin ^a , grapefruit juice, ketoconazole
CYP3A5	Thought to be same as CYP3A4
CYP3A7	Unclear at this time but may be similar to CYP3A4

Phase I:

- **Phase I reactions not involving the P450 system:**
- These include
 - ✓ amine oxidation (for example, oxidation of catecholamines or histamine),
 - ✓ alcohol dehydrogenation (for example, ethanol oxidation),
 - ✓ esterases (for example, metabolism of *pravastatin* in liver),
 - ✓ hydrolysis (for example, of *procaine*).

Phase - II reactions (conjugation reactions):

- This is synthetic process by which a drug or its metabolite is combined with an endogenous substance resulting in various conjugates
- Drugs which contain
 - ✓ a) Hydroxyl
 - ✓ b) Sulfate conjugation
 - ✓ c) Acetyl conjugation
 - ✓ d) Glycine conjugation
 - ✓ e) Methylation

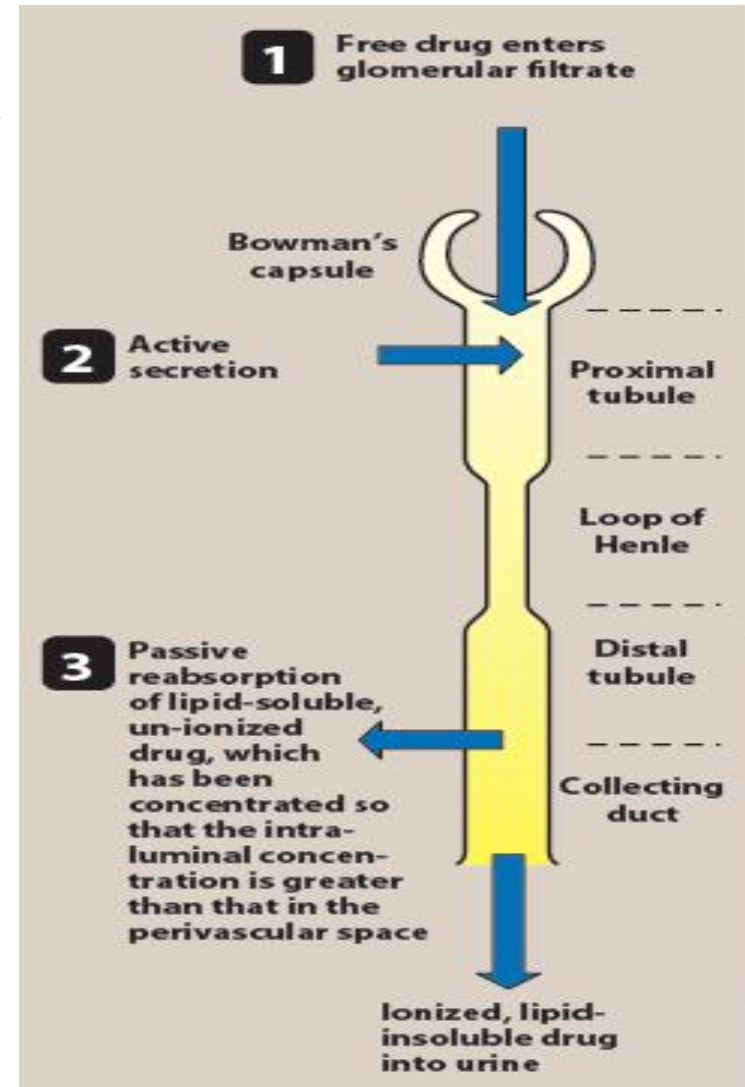


Excretion of drugs

- Excretion of drugs means the transportation of unaltered or altered form of drug out of the body.
- The major processes of excretion include
 - ✓ renal excretion,
 - ✓ hepatobiliary excretion
 - ✓ pulmonary excretion.
- The minor routes of excretion are
 - ✓ saliva, sweat, tears, breast milk, vaginal fluid, and hair.

Renal excretion

- kidney is the primary organ of removal for most drugs , especially for those that are **water soluble** and **not volatile**.
- The three principal processes that determine the urinary excretion of a drug are
 - ✓ glomerular filtration,
 - ✓ tubular secretion,
 - ✓ tubular reabsorption



- **1.Glomerular filtration:**
- Drugs enter the kidney through renal arteries,
- Free drug (not bound to albumin) flows through the capillary slits into Bowman's space as part of the glomerular filtrate
- Lipid solubility and pH do not influence the passage of drugs into the glomerular filtrate.
 - ✓ varying the glomerular filtration rate and plasma binding of the drugs may affect this process

- **2. Proximal tubular secretion:**
- Drugs were not transferred into the glomerular filtrate leave the glomeruli through efferent arterioles,
- Secretion primarily occurs in the proximal tubules by two energy-requiring active transport systems
- Each of these transport systems shows low specificity and can transport many compounds.

- **3. Distal tubular reabsorption:**
- drug concentration increases and exceeds that of the perivascular space.
- uncharged, may diffuse out of the nephric lumen, back into the systemic circulation.
- ionized form of the drug in the lumen may be done to minimize the amount of back-diffusion
 - ✓ weak acids can be eliminated by alkalization of the urine,
 - ✓ weak bases may be increased by acidification of the urine.

- The total body (systemic) clearance, CL_{total} is the sum of the clearances from the various drug-metabolizing and drug-eliminating organs.
- The kidney is often the major organ of excretion; however, the liver also contributes to drug loss through metabolism and/or excretion into the bile.
- Some drugs may also be reabsorbed through the **enterohepatic circulation**, thus prolonging their half-lives.
- Total clearance can be calculated by using the following equation:
 - ✓ $CL_{total} = CL_{hepatic} + CL_{renal} + CL_{pulmonary} + CL_{other}$
 - ✓ where $CL_{hepatic} + CL_{renal}$ are typically the most important.

Hepatobiliary excretion

- the conjugated drugs are excreted by hepatocytes in the bile.
 - ✓ polar drugs
- **Back up pathway** when renal function is impaired.
- After excretion of drug through bile into intestine, certain amount of drug is **reabsorbed** into portal vein leading
 - ✓ can prolong the action of drug
 - ✓ e.g. chloramphenicol, oral estrogen
- Tetracyclines excreted by biliary tract ,used for treatment of biliary tract infection.

Half-lives and Kinetics

- Half-life:
- Plasma half-life: Time it takes for plasma concentration of a drug to drop to 50% of initial level.
- Whole body half-life: Time it takes to eliminate half of the body content of a drug.

Factors affecting half-life

- age
- renal excretion
- liver metabolism
- protein binding

Half-lives and Kinetics

- The half-life of a drug is increased by
 - ✓ 1) diminished renal plasma flow or hepatic blood flow,
 - ✓ 2) decreased ability to extract drug from plasma,
 - ✓ 3) decreased metabolism,
- On the other hand, the half-life of a drug may decrease by
 - ✓ 1) increased hepatic blood flow,
 - ✓ 2) decreased protein binding,
 - ✓ 3) increased metabolism.

Order of kinetics

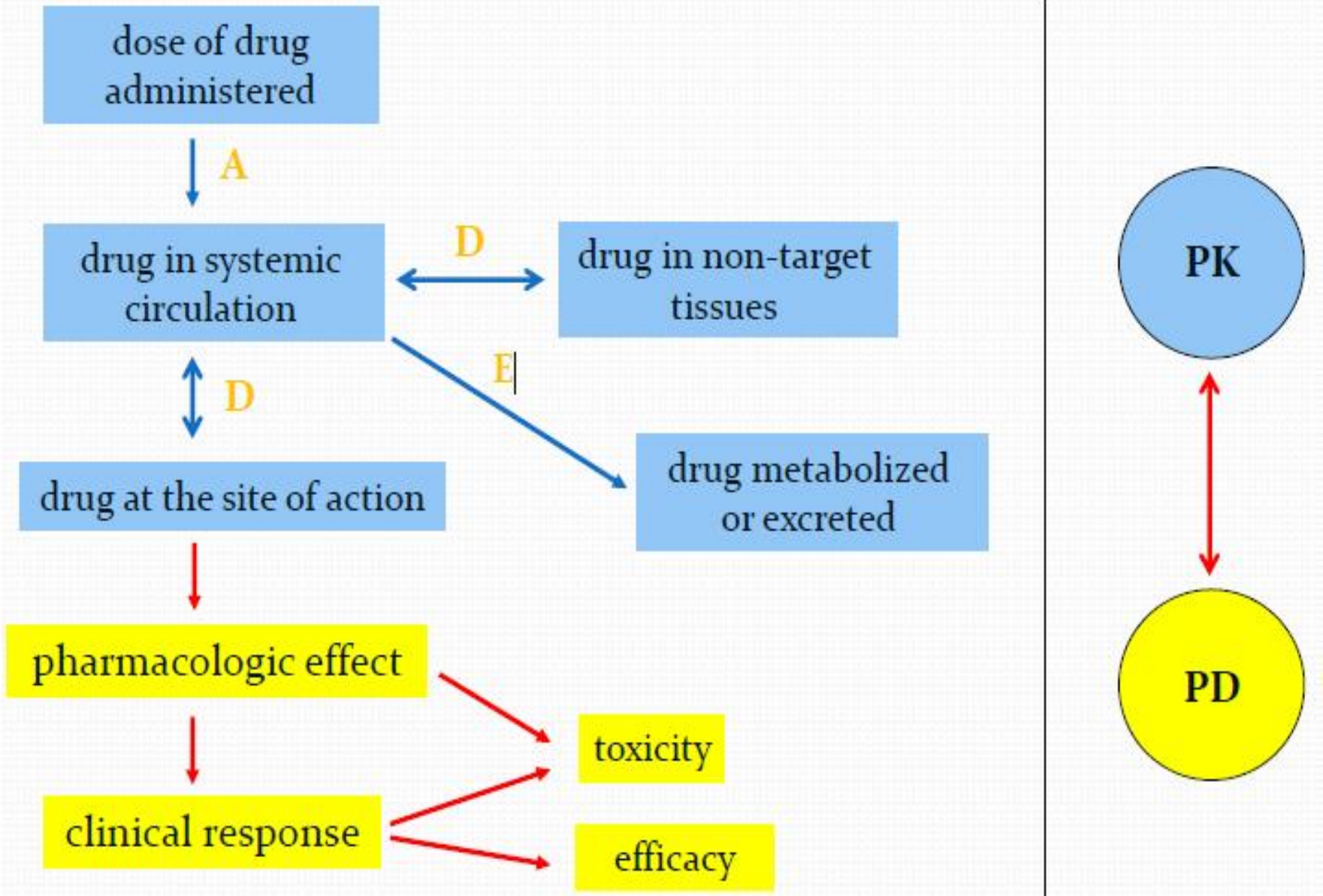
- **First order:**
- This is the most common process for many drugs.
- The rate at which absorption, distribution, metabolism and excretion occur **proportional** to the concentration of drugs
- i.e. constant fraction of the drug in each equal interval of time.
- A patient was given 100mg drug orally by proportion of 10% per minute
- At First minute $10\% = 10\text{gm}$
- At Second minute $10\% = 9\text{gm}$
- At third minute $10\% = 8.1\text{gm}$

- **Zero order kinetic:**
- It is **independent of the amount** of drug present at the particular sites of drug absorption or elimination.
- Few drugs follow this process e.g. ethanol, phenytoin.
- Here constant amount of the drug is eliminated in each equal interval of time.
- On repeated administration of drug after certain stage it goes on accumulating in the body and leads to toxic reactions.
- E.g. A patient given 100mg drug orally by proportion of 10mg/minute, absorption or elimination is?
- At First minute = 10gm
- At Second minute = 10gm
- At third minute = 10gm

Pharmacodynamics

- Pharmacodynamics deals with the study of
 - ✓ the biochemical and physiological effects of drugs
 - ✓ their mechanisms of action.
 - ✓ complete action-effect sequence and the dose-effect relationship.
 - ✓ Modification of the action of one drug by another drug is also an aspect of pharmacodynamics.
 - ✓ **In general what the drug does to the body**

What Happens After Drug Administration?



Mechanisms of drug action

1.Receptor involving mechanism

- ✓ the drugs act by interacting with a cellular component called receptor
- ✓ This is the mechanism of for most drugs e.g. adrenergic receptors, cholinceptors, insulin receptors

2. non-receptor mechanisms:

- ✓ Some drugs act through simple physical or chemical reactions without interacting with any receptor. Eg antacids, charcoal

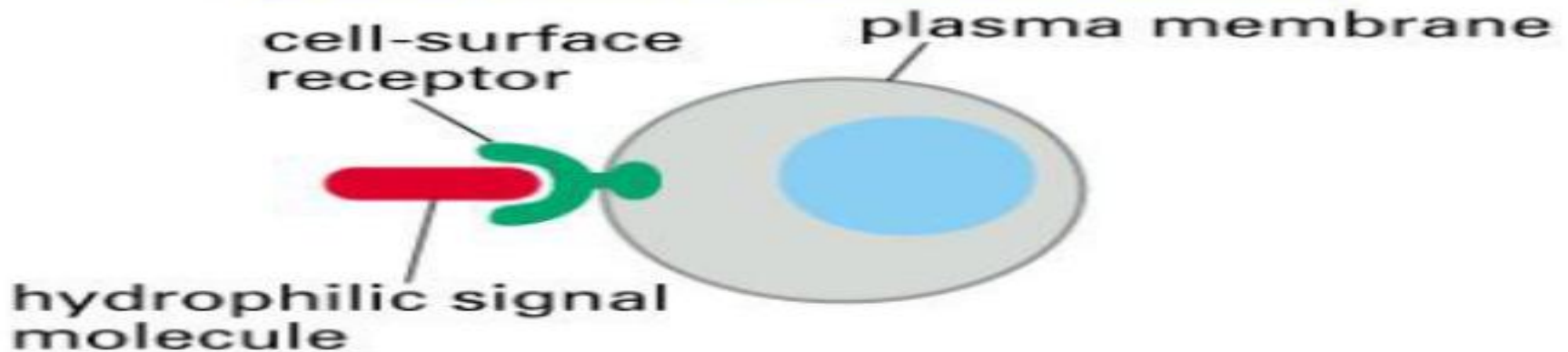
■ Receptor:-

- a macromolecule typically made of proteins that interacts with either:
 - ✓ endogenous ligand or
 - ✓ drug to mediate a pharmacological or physiological effect
- Cells may have different types of receptors, each of which is specific for a particular ligand

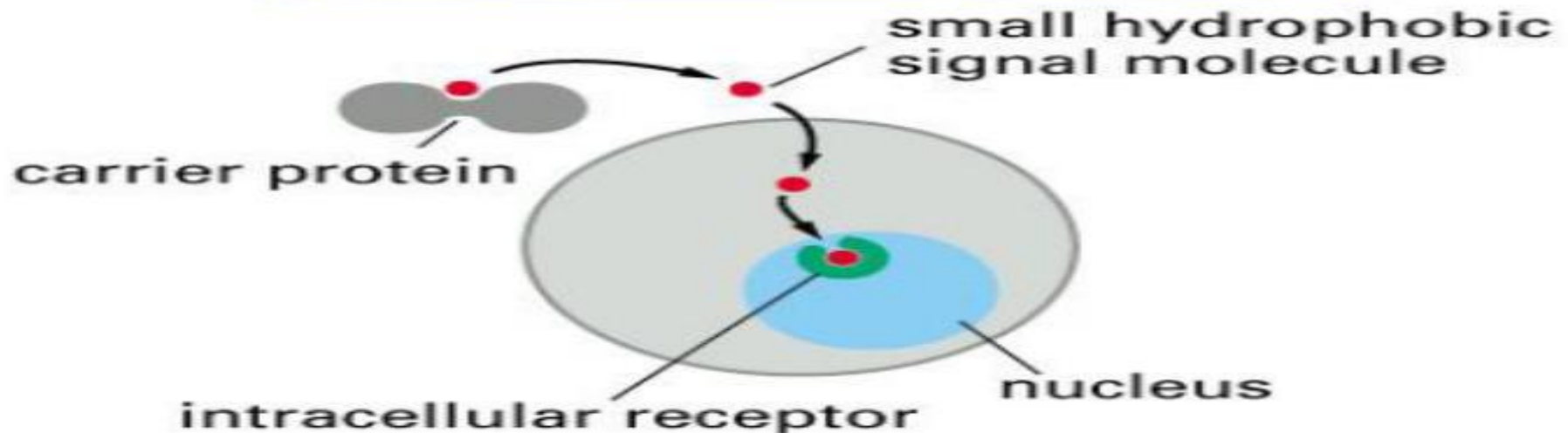
Drug + Receptor $\rightleftharpoons$ Drug-receptor complex $\longrightarrow$ Biological effect

Receptors present either on the cell surface or with in the cell

CELL-SURFACE RECEPTORS



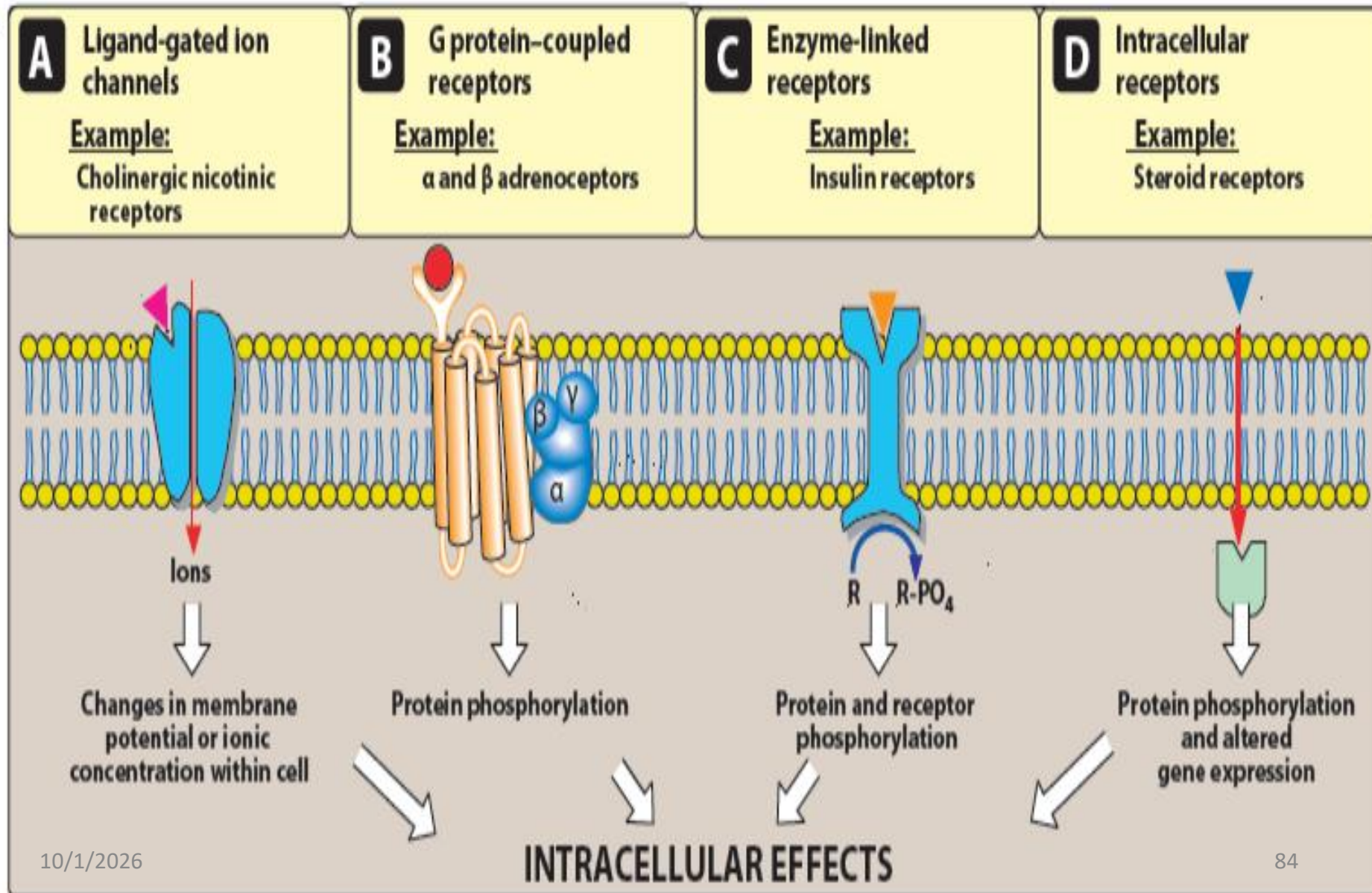
INTRACELLULAR RECEPTORS



Major Receptor Families

- enzymes and structural proteins can be considered to be pharmacologic receptors
- proteins are the richest sources of therapeutically exploitable pharmacologic receptors
- These receptors may be divided into four families:
 1. ligand-gated ion channels,
 2. G protein coupled receptors,
 3. enzyme-linked receptors, and
 4. intracellular receptors

Trans membrane signaling mechanisms



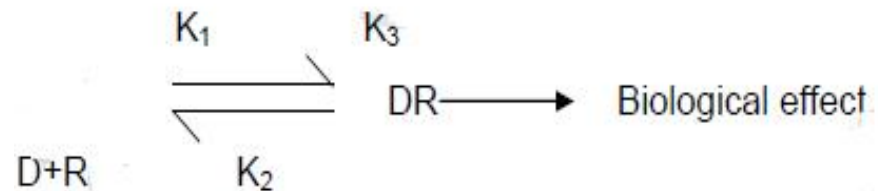
G protein coupled receptors

- **first messengers** like neurotransmitters, hormones, autacoids and most of drugs bind with their specific receptors,
 - ✓ the drug receptor complex is formed
- which causes the **synthesis** and release of another intracellular regulatory molecule termed as **second messengers**
- e.g. cyclic AMP, calcium, cyclic GMP, inositol triphosphate (IP3), diacylglycerol and calmodulin

activation of adenylyl cyclase → cAMP → activation of kinases → protein phosphorylation → eg opening of some calcium channels as well as in the activation of other enzymes

Drug- receptor bonds

- The forces that attract the drug to its receptor are termed chemical bonds and they are
 - (a) hydrogen bond
 - (b) ionic bond
 - (c) **covalent bond**
 - (d) Vander waals force.
 - Covalent bond is the strongest bond
- ✓ drug-receptor complex is usually **irreversible**.



Where D = Drug, R= receptor DR= Drug receptor complex (affinity)

K_1 = association constant

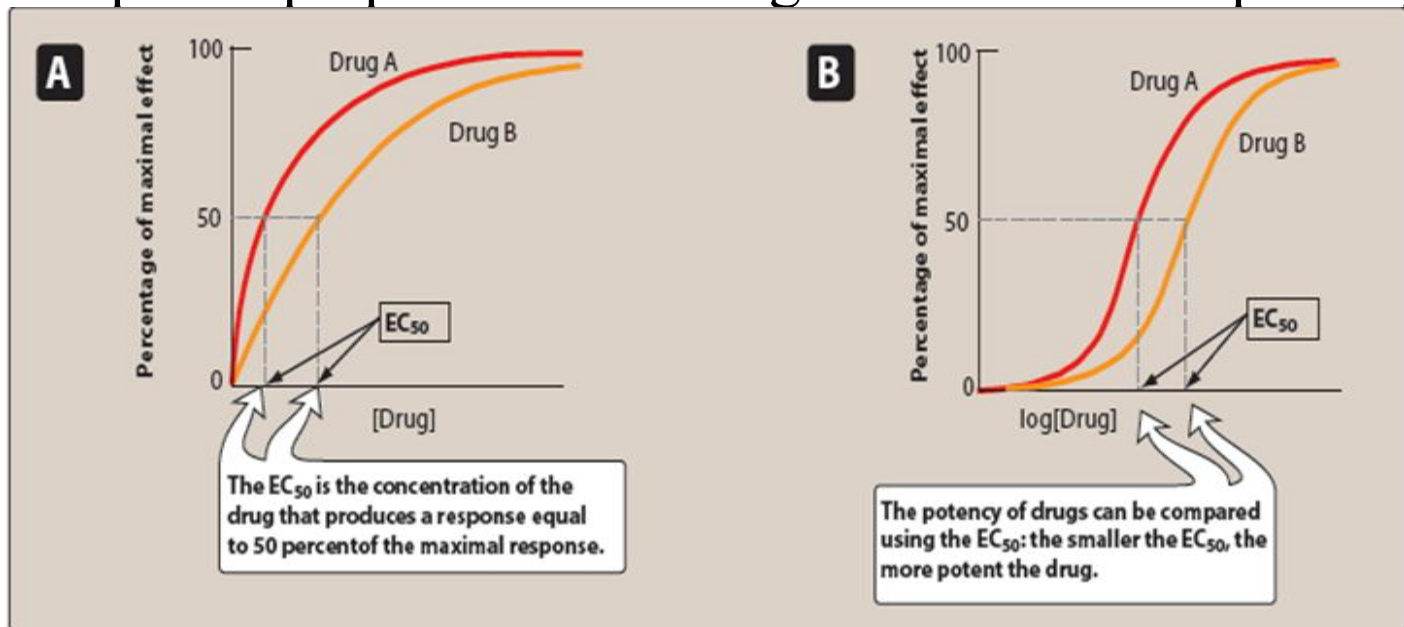
K_2 = dissociation constant

K_3 = intrinsic activity

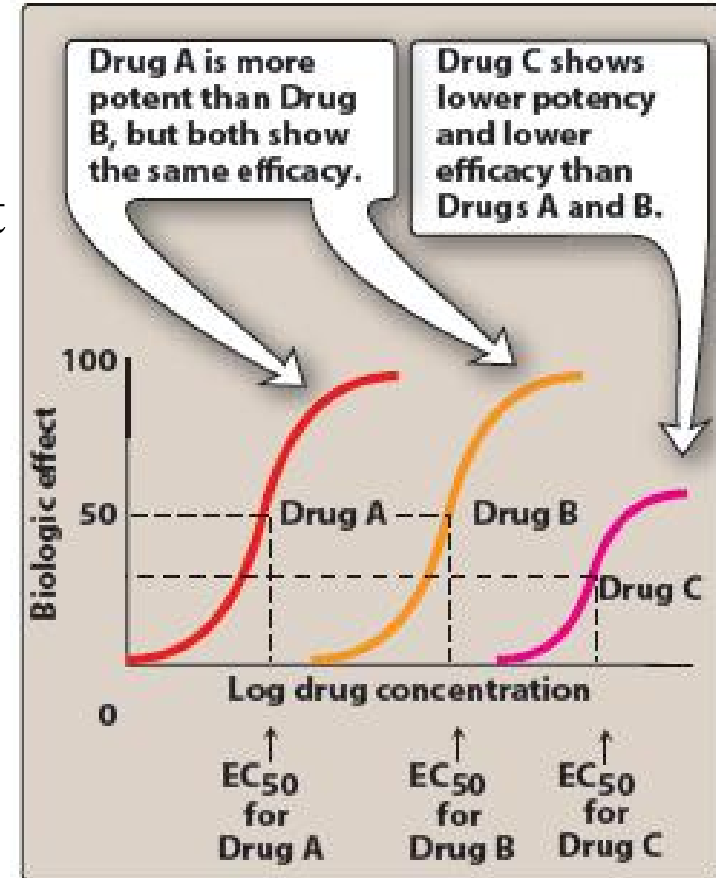
Graded dose–response relations

- As the concentration of a drug increases, the magnitude of its pharmacologic effect also increases.
- The response is a graded effect, meaning that the response is continuous and gradual.
- Two important properties of drugs, **potency** and **efficacy**, can be determined by graded **dose–response curves**.

- **Potency:** is a measure of the amount of drug necessary to produce an effect of a given magnitude.
- The concentration of drug producing an effect that is 50 percent of the maximum is used to determine potency and is commonly designated as the EC₅₀.
- Thus, therapeutic preparations of drugs will reflect the potency.



- **Efficacy**(intrinsic activity): is the ability of a drug to elicit a response when it interacts with a receptor.
- efficacy, is more important than drug potency.
- A drug with greater efficacy is more therapeutically beneficial than one that is more potent.
- Maximal efficacy of a drug assumes that
 - ✓ all receptors are occupied by the drug,
 - ✓ and no increase in response
 - if more drugs are added.
- **Affinity**: the ability to bind with the receptor

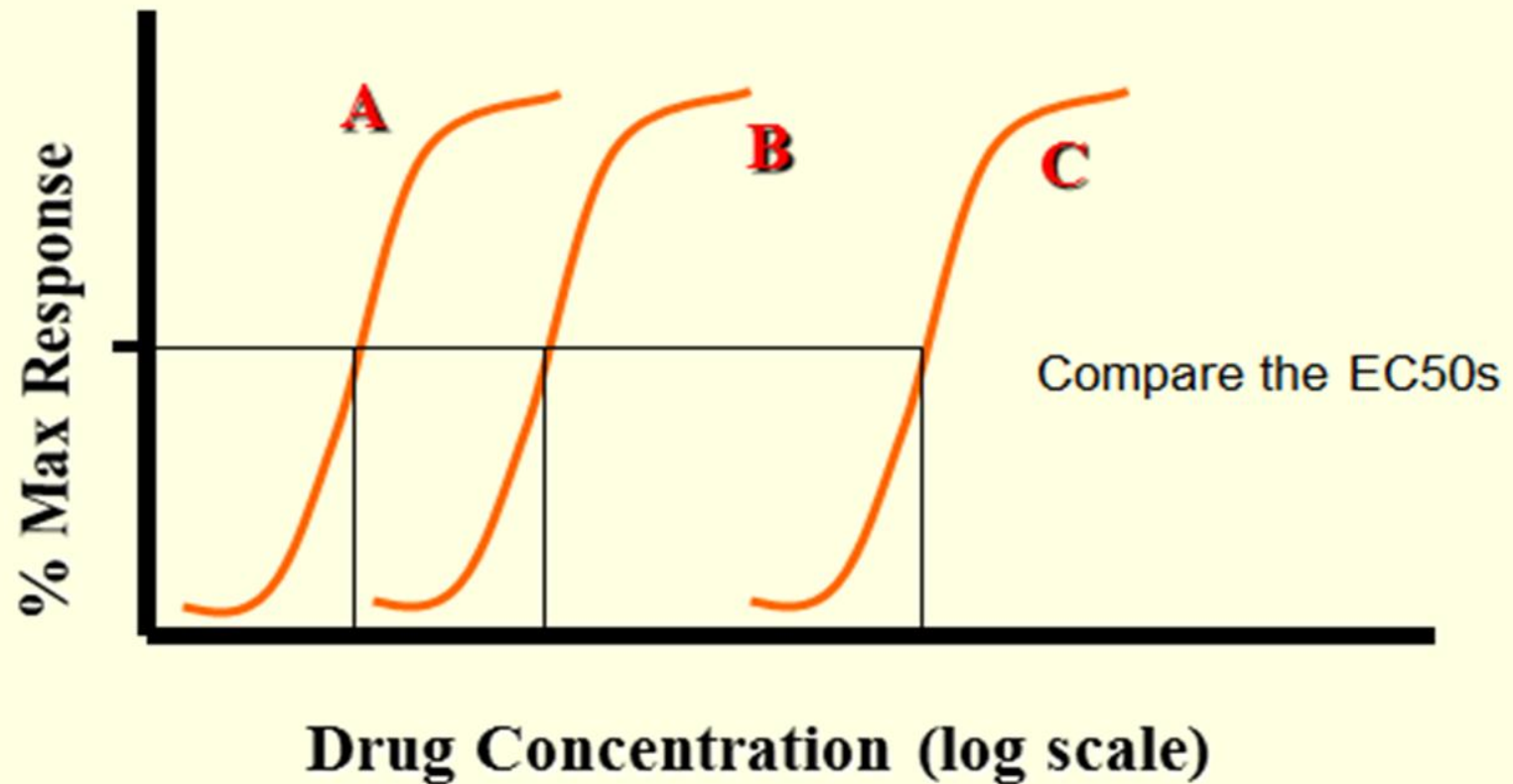


Drug-receptor interaction

Agonists

- An agonist binds to a receptor and produces a biologic response.
- An agonist may mimic the response of the endogenous ligand on the receptor,
- **Full agonists:** a drug binds to a receptor and produces a maximal biologic response that mimics the response to the endogenous ligand,
- **Partial agonists:** have efficacies greater than zero but less than that of a full agonist
- **Inverse agonist:** An agent which activates a receptor to produce an effect in the opposite direction to that of the agonist

Agonists



Full Agonists (i.e., **equal efficacies**) that
Differ In **Potency**:

Antagonists

- Antagonists are drugs that decrease or oppose the actions of another drug or endogenous ligand.
- An antagonist has no effect if an agonist is not present, no effect by themselves.
- types of antagonism can be classified
 - 1. Chemical antagonism
 - 2. Functional antagonism
 - 3. Competitive antagonism
 - **a.** Equilibrium competitive
 - **b.** Nonequilibrium competitive
 - 4. Noncompetitive antagonism

❖ Chemical Antagonism

- involves a direct chemical interaction between the agonist and antagonist in such a way as to render the agonist pharmacologically inactive.
- E.g. chelating agents for biological inactivation and removal from the body of toxic metals., dimercaprol, is used in the treatment of toxicity from mercury, arsenic, and gold.

❖ Functional Antagonism

- ✓ may act at a completely separate receptor, initiating effects that are functionally opposite effects on several body functions.
- ✓ E.g. epinephrine to histamine induced bronchoconstriction
- ✓ acetylcholine and epinephrine, heart, intestinal movement, pupil

- **Competitive Antagonism**

- ✓ is the most frequently encountered in clinical practice.
- ✓ antagonist & agonist combines with the same site on the receptor, competes with this binding site.
- ✓ fall into either of two subtypes, depending on the type of bond formed between the antagonist and the receptor.
- ✓ equilibrium competitive or reversibly competitive.
 - If the bond is a loose one, the antagonism
- ✓ Non equilibrium competitive or irreversibly competitive.
 - bond is covalent
 - is not readily reversible

❖ **Noncompetitive Antagonism**

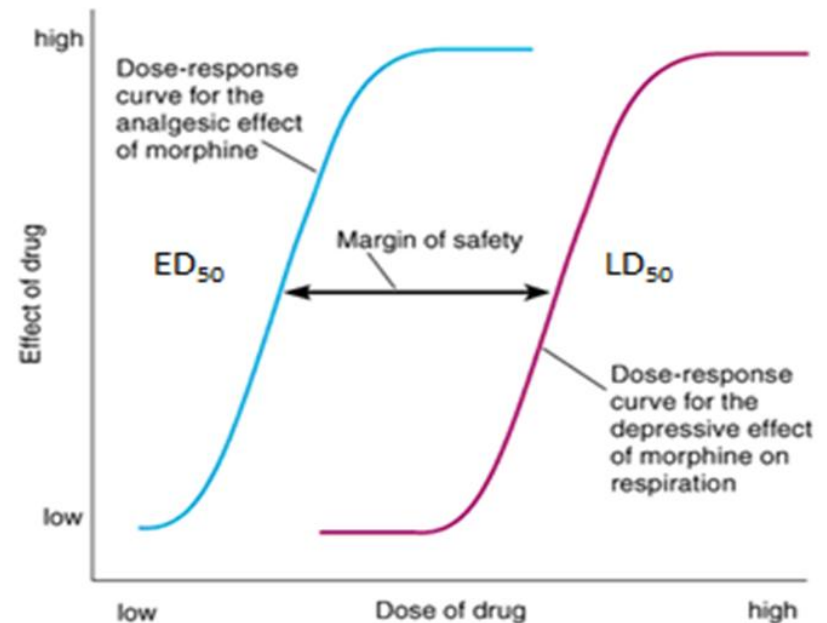
- ✓ The inhibitor reacts with an adjacent site and not with the catalytic site, but alters the enzyme in such a way that it loses its catalytic property.
- Agonists have both affinity and maximal intrinsic activity, e.g. adrenaline, histamine, morphine.
- Competitive antagonists have affinity but no intrinsic activity, e.g. propranolol, atropine, chlorpheniramine, naloxone.
- Partial agonists have affinity and submaximal intrinsic activity e.g. pentazocine
- Inverse agonists have affinity but intrinsic activity with a minus s

Quantal dose effect

- It is all or none response, the sensitive objects give response to **small doses** of a drug while some will be **resistant** and need very large doses.
- The quantal dose effect curve is often characterized by stating the **median effective dose** and the **median lethal dose**.
- **Median lethal dose or LD50:** This is the dose (mg/kg), which would be expected to kill half of a population of the same species and strain.
- **Median effective dose or ED50:** This is the dose (mg/kg), which produces a desired response in 50 per cent of test population.

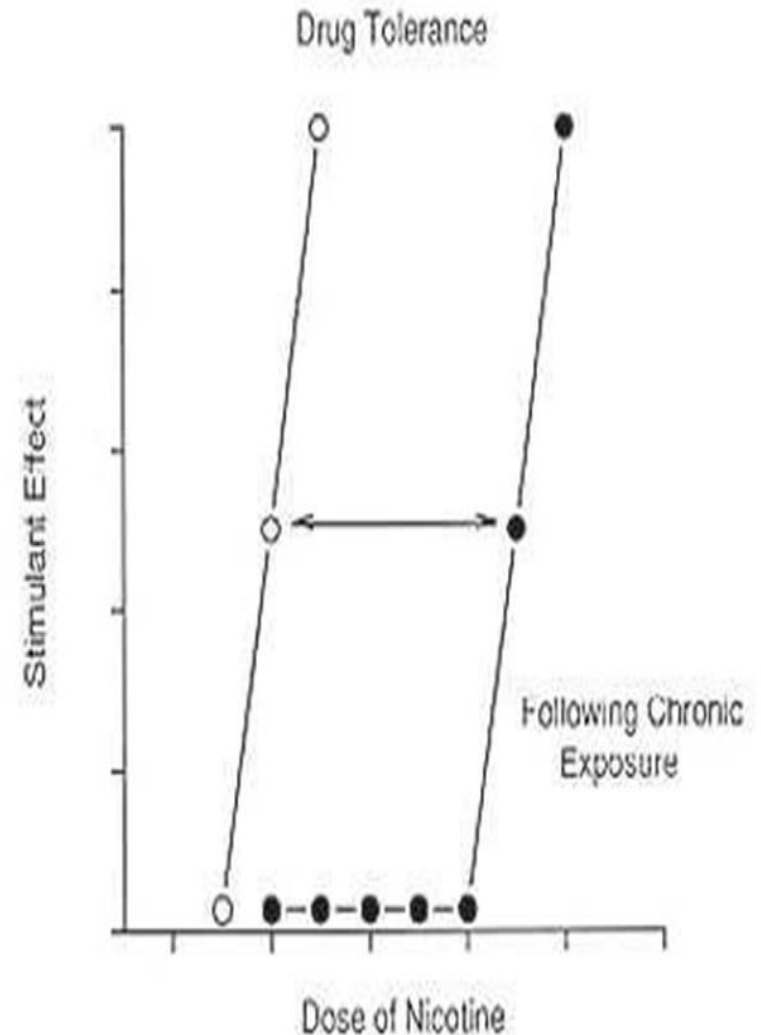
Therapeutic Index(TI)

- It is an approximate assessment of the **safety of the drug**.
- It is the ratio of the median lethal dose and the median effective dose.
- Also called as therapeutic window or safety or margin of safety.
- $TI = LD_{50}/ED_{50}$, an indication of safety of a drug (higher is better)



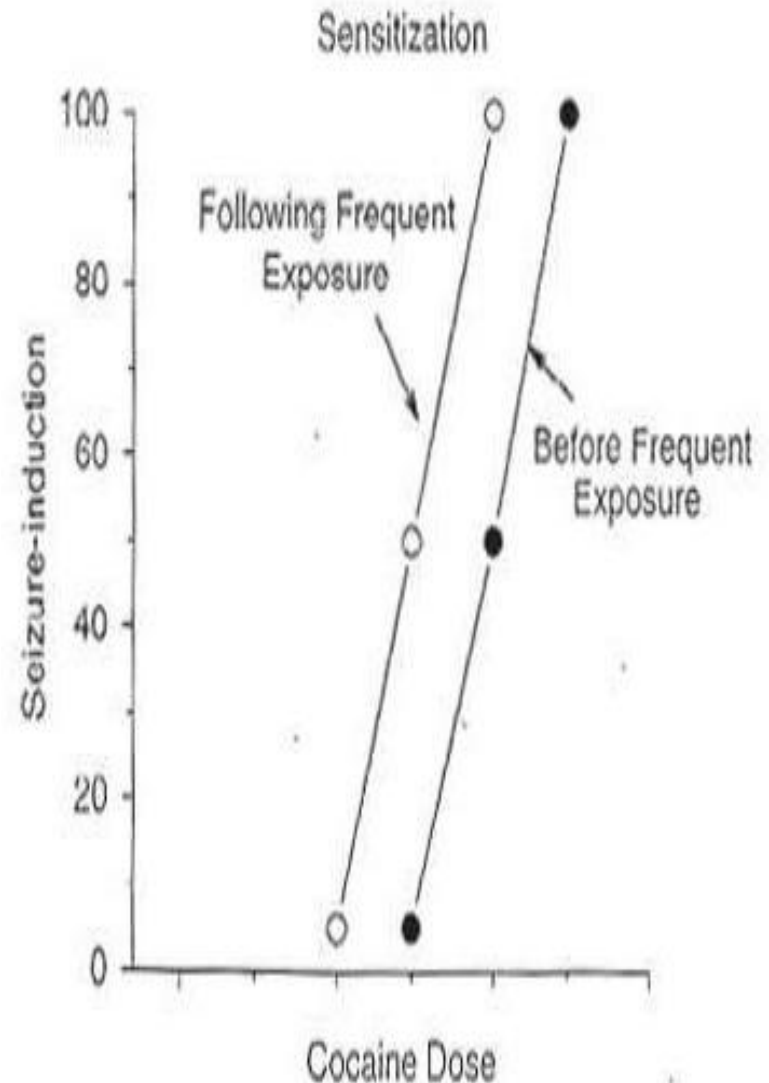
Tolerance(desensitization)

- Decreased response to same dose with repeated (**constant**) exposure *or* more drug needed to achieve same effect
- Right-ward shift of D-R curve Sometimes occurs in an acute dose (e.g. alcohol)
- Can develop across drugs (cross-tolerance)
- Caused by compensatory mechanisms that oppose the effects of the drug



Sensitization

- Increased response to same dose with repeated (**binge-like**) exposure
- *or* less drug needed to achieve same effect
- Left-ward shift in D-R curve
- Sometimes occurs in an acute dose (e.g. amphetamine)
- Can develop across drugs (cross-sensitization)



Mechanisms of Tolerance and Sensitization

- **Pharmacokinetic**

- ✓ changes in drug availability at site of action (decreased bioavailability)
- ✓ Decreased absorption
- ✓ Increased binding to depot sites

- **Pharmacodynamic**

- ✓ changes in drug-receptor interaction
- ✓ G-protein uncoupling
- ✓ Down regulation of receptors

- **Psychological**

- Set (expectations) and setting (environment)

- **Motivational**

- **Habituation**

- automatic physiological change in response to cues

- **Metabolic**

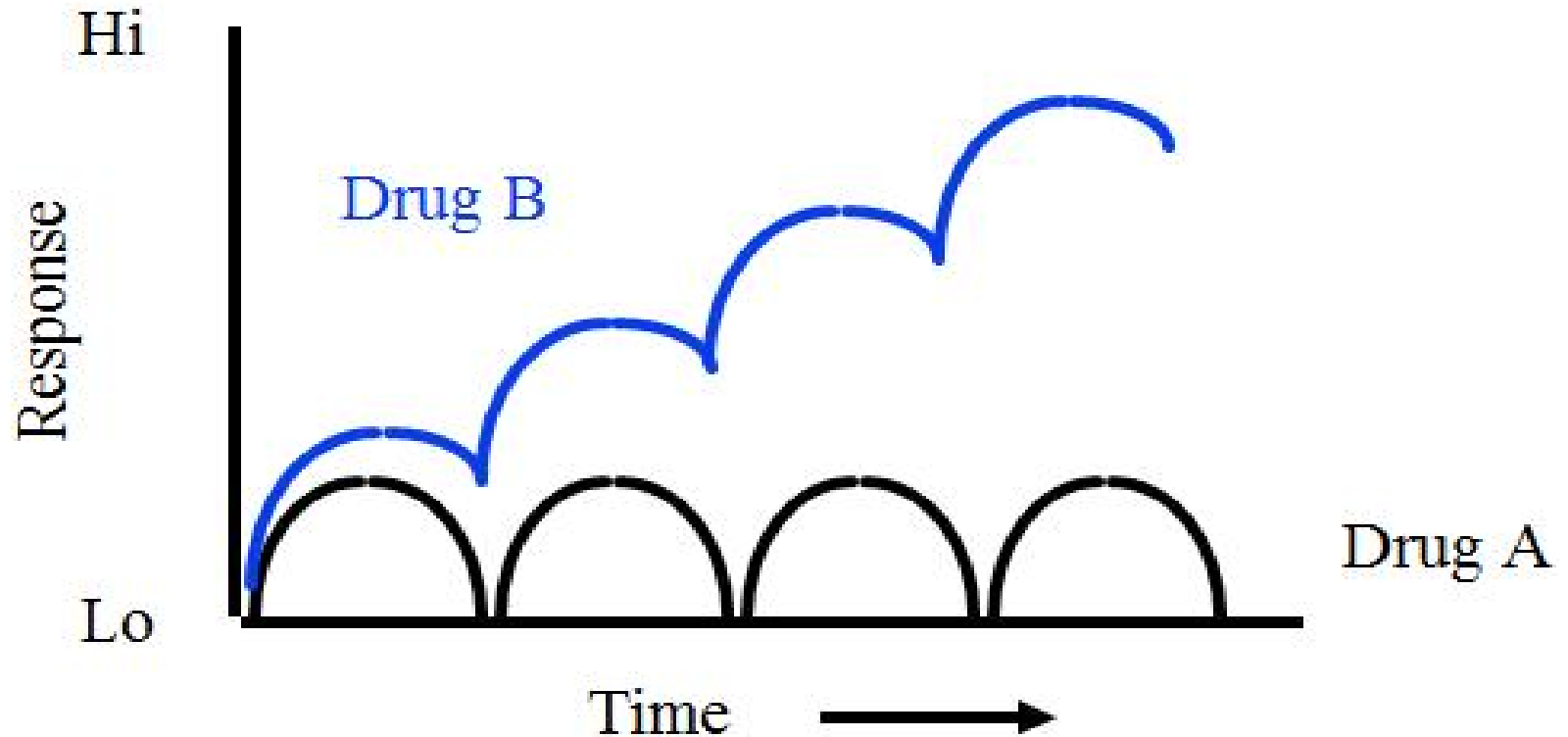
- break down and/or excrete the drug more quickly due to repeated exposure.

—————> Increased excretion

Drug-drug Interactions

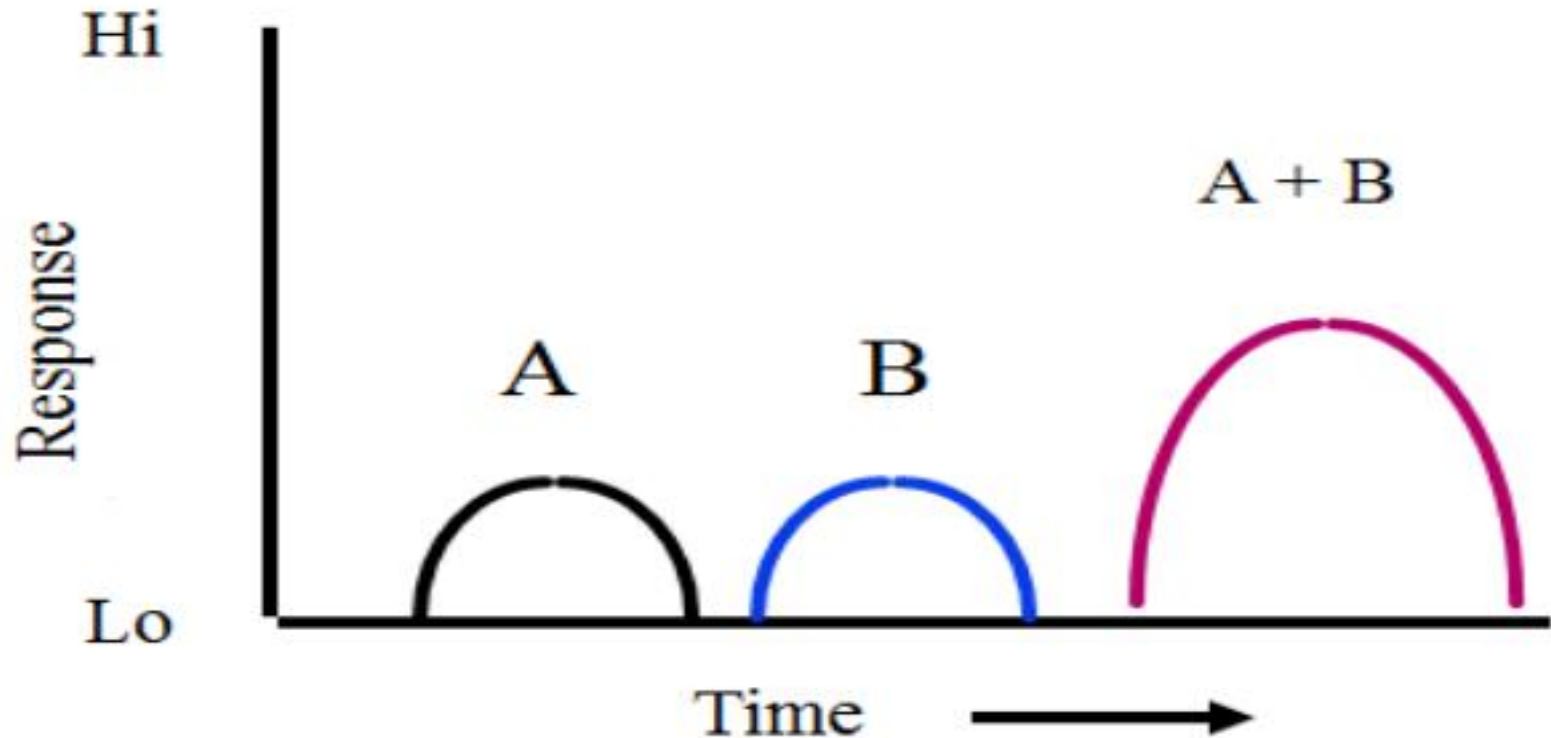
- Pharmacokinetic and pharmacodynamics
 - ✓ With pharmacokinetic drug interactions, one drug affects the absorption, distribution, metabolism, or excretion of another.
 - ✓ With pharmacodynamics drug interactions, two drugs have interactive effects in the brain.
 - ✓ Either type of drug interaction can result in adverse effects in some individuals.
 - ✓ In terms of efficacy, there can be several types of interactions between medications: **cumulative, additive, synergistic, and antagonistic.**

Cumulative Effects



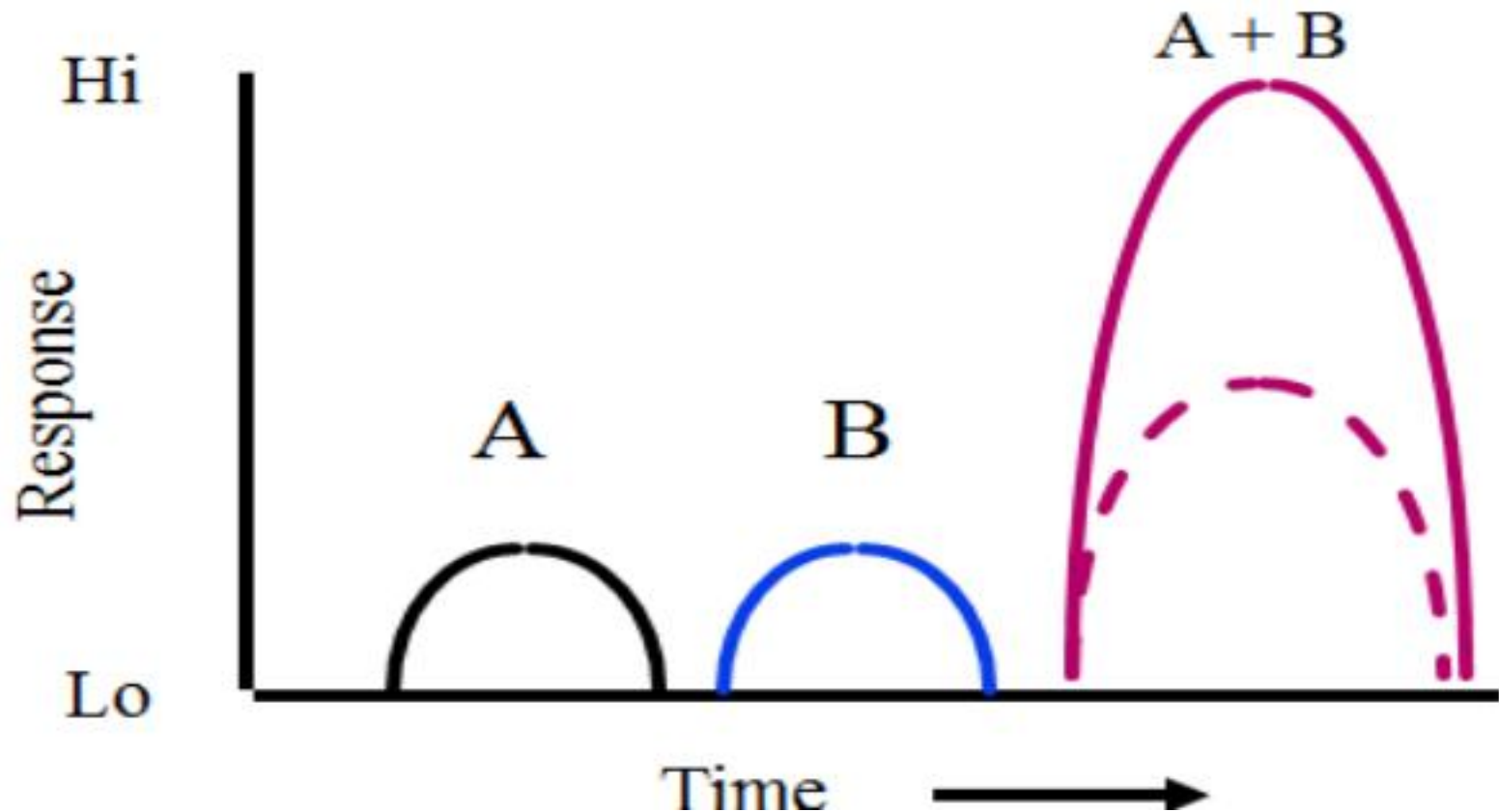
The condition in which repeated administration of a drug may produce effects that are more pronounced than those produced by the

Additive Effects



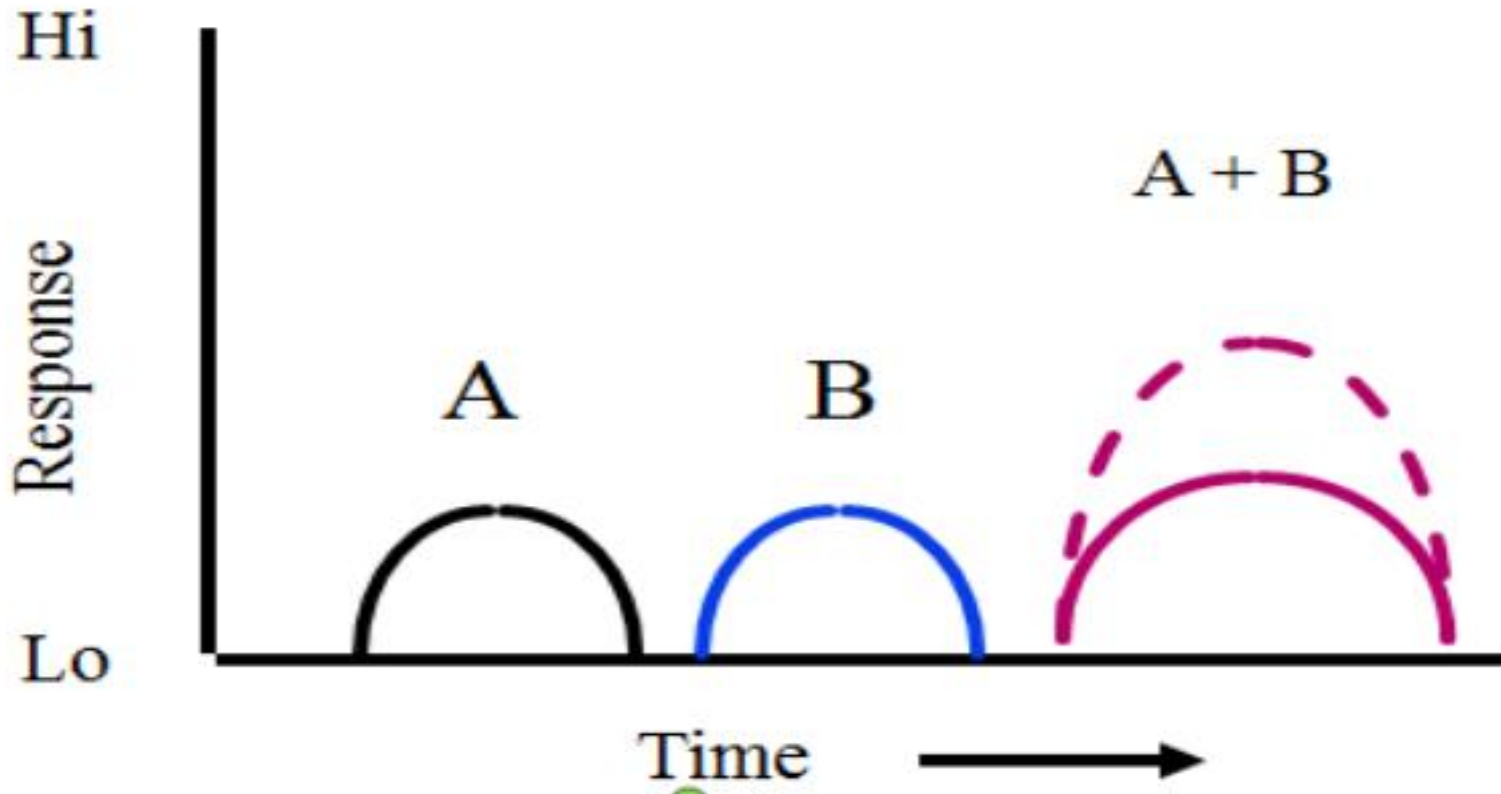
- ❖ The effect of two chemicals is equal to:-
 - ✓ the sum of the effect of the two chemicals taken separately.

Synergistic Effects



- The effect of two chemicals taken together is greater than the sum of their separate effect at the same doses, e.g., alcohol and other drugs

Antagonistic Effects



- ❖ The effect of two chemicals taken together is less than the sum of their separate effect at the same doses

ADVERSE DRUG REACTION

Any noxious change which is

- ✓ Suspected to be due to a drug
- ✓ At **doses normally** used in man
- ✓ May requires treatment or decrease in dose or
- ✓ Caution in the future use of the same drug



- **Incidence of ADR more**
 - Polypharmacy
 - Elderly
 - Children
 - Patient with multiple diseases
 - Pregnancy
 - Malnourished
 - Immunosuppression
 - Drug Abusers and addicts

- **Develop**
 - ✓ Immediately
or
 - ✓ Prolonged medication
or
 - ✓ After stopping.

GRADING OF SEVERITY OF ADVERSE DRUG REACTIONS :

- **Minor** : No therapy, antidote or prolongation of hospitalization is required.
- **Moderate**: Requires change in drug therapy, specific treatment or prolongs hospital stay.
- **Severe**: **Potentially** life-threatening, causes permanent damage or requires intensive medical treatment.
- **Lethal** : Directly or indirectly contributes to death of the patient.

CLASSIFICATIONS OF ADR

- **A (Augmented)**
- **B (Bizarre)**
- **C (Continuous)**
- **D (Delayed)**
- **E (Ending Use)**
- **F (Failure of Efficacy)**

Broadly

Type- A (Predictable)- Based on pharmacological properties

Type- B (Non-predictable) – Based on Immunological response
and genetic makeup of person

TYPE A- AUGMENTED

- These are based on the pharmacological properties of the drug so can be predicted.
- They are common and account for 75% of ADRs
- Dose related and preventable mostly reversible.

Examples:-

- Anticoagulants (e.g., warfarin, heparin) – bleeding
- Anti-hypertensives (e.g.. α 1-antagonists) – hypotension
- Anti-diabetics (e.g. insulin) - hypoglycemia

TYPE B- BIZZARE OR UNPREDICTABLE

- Have **no direct relationship** to the dose of the drug or the pharmacological mechanism of drug action.
- Develop on the basis of:
 - Immunological reaction on a drug (**Allergy**)
 - Genetic predisposition (**Idiosyncratic reactions**)
- More serious clinical outcomes with higher mortality and morbidity.
- Mostly require immediate withdrawal of the drug.

TYPE C – CHRONIC (CONTINUOUS) USE

- They are mostly associated with **cumulative-long term** exposure

Example:-

Analgesic (NSAID)– interstitial nephritis, papillary sclerosis, necrosis

TYPE D – DELAYED

- They manifest themselves with significant delay
 - **Teratogenesis** -Thalidomide – Phocomelia (flipper-like fore limbs)
 - **Mutagenesis/Cancerogenesis**

TYPE E – END OF USE

- Drug withdrawal syndromes and rebound phenomenons
 - Example – sudden withdrawal of long term therapy with β -blockers can induce rebound tachycardia and hypertension

PHARMACOVIGILANCE (DAUP)

- The 'science and activities relating to the **d**etection, **a**ssessment, **u**nderstanding and **p**revention of adverse effects or any other drug related problems'
- The information generated is useful in educating doctors and in the official regulation of drug use.
- It has an important role in **rational use** of medicines, as it provides the basis for assessing **safety** of medicines.

Various activities involved in pharmacovigilance are:

- **Postmarketing surveillance** and other methods of ADR monitoring such as voluntary reporting by doctors prescription event monitoring.
- **Dissemination of ADR data** through 'drug alerts', 'medical letters,' advisories sent to doctors by pharmaceuticals and regulatory agencies.
- **Changes in the labelling** of medicines indicating restrictions in use or statutory warnings, precautions, or even withdrawal of the drug.

Categorized into:

- Side effects-
- Secondary effects
- Toxic effects
- **Intolerance**
- **Idiosyncrasy**
- Drug allergy
- Photosensitivity
- Drug dependence
- Drug withdrawal reactions
- Teratogenicity
- Mutagenicity and Carcinogenicity
- Drug induced diseases (**latrogenic** disorders or latrogenicity)

Beware of – latrogenic, Idiosyncrasy, Idiopathic, Intolerance

SIDE EFFECTS

- Unwanted often unavoidable Pharmacodynamic effects.
- Occur at therapeutic doses.
- Predictable

Examples. H1 Anti-histaminics- Sedation

An effect may be therapeutic in one context but side effect in another context

- Constipation by codeine is side effect but can be used as therapeutic effect in patient with loose motions

SECONDARY EFFECTS

- Indirect consequences of a primary action of the drug.
- E.g. corticosteroids weaken host defence mechanisms so that latent tuberculosis gets activated.

TOXIC EFFECTS (Poisonous effect)

- Over dose or prolonged use.
- The effects are predictable and dose related.
- The CNS, CVS, kidney, liver, lung, skin and bone marrow are most commonly involved in drug toxicity.
- Toxicity may result from extension of the therapeutic effect itself, e.g. complete A-V block by digoxin, bleeding due to heparin.
- **Poisoning:** Poison is a substance which endangers life by severely affecting one or more vital functions.

INTOLERANCE

- It is the appearance of characteristic toxic effects of a drug in an individual at therapeutic doses
- It indicates a low threshold of the individual to the action of a drug
- **Example:-** Only few doses of carbamazepine may cause ataxia in some people

IDIOSYNCRASY

- It is **genetically determined abnormal reactivity** to a chemical.
- The drug interacts with some unique **feature of the individual**, not found in majority of subjects, and produces the uncharacteristic reaction.

Example :-Chloramphenicol produces non dose-related serious aplastic anaemia in rare individuals.

- Barbiturates cause excitement and mental confusion in some individuals

DRUG ALLERGY

- It is also called **drug hypersensitivity**.
- It is an **immunologically mediated reaction** producing stereotype symptoms which are unrelated to the pharmacodynamic profile of the drug.
- It generally occur even with **much smaller doses** and have a different time course of onset and duration.

- Allergic reactions occur only in a **small proportion of the population** exposed to the drug .
- History of prior sensitization may or may not be evident.
- The drug or its metabolite acts as antigen (AG) or more commonly **hapten** (incomplete antigen) and induce production of antibody (AB)/sensitized lymphocytes.

TYPES OF ALLERGIC REACTIONS

A) HUMORAL

1. **Type I/ anaphylactic reactions.**
2. **Type-II / cytolytic reactions.**
3. **Type-III / retarded or Arthus reactions.**

B) CELL MEDIATED

Type-IV (delayed hypersensitivity) reactions.

PHOTOSENSITIVITY

- It is a cutaneous reaction resulting from drug induced sensitization of the skin to **UV radiation**.
- The reactions are of two types:
 - a) **Photo-toxic :- (T-S)**
 - a) Drug or its metabolite **Accumulates** in the skin,
 - b) absorbs light and undergoes a **Photochemical reaction** followed by
 - c) **Photobiological reaction** resulting in
 - d) Tissue damage (sunburn-like),
 - a) i.e. erythema, edema, blistering , hyper pigmentation, desquamation.

The shorter wave lengths (290-320 nm, UVB) are responsible

(b) Photo-allergic: (A-L)

- ✓ Drug or its metabolites induce a cell mediated immune response which on exposure to Light of **longer wave lengths (320-400 nm, UV -A)**
- ✓ Produces a papular or eczematous contact dermatitis like picture.
- ✓ Drugs involved are sulfonamides, sulfonylureas, griseofulvin, chloroquine, chlorpromazine

DRUG DEPENDENCE

- Use of drugs for personal satisfaction
- Higher priority than other basic needs, often in the face of known risks to health.

Physical dependence It is an altered physiological state produced by repeated administration of a drug which necessitates the continued presence of the drug to maintain physiological equilibrium.

- Discontinuation of the drug results in a characteristic **withdrawal (abstinence) syndrome.**
- Drugs producing physical dependence are opioids, barbiturates and other depressants including alcohol and benzodiazepines

- **Drug abuse :**
- Refers to use of a drug by self medication in a manner and amount that deviates from the approved medical and social patterns in a given culture at a given time.
- **Drug addiction**
- It is a pattern of compulsive drug use characterized by overwhelming involvement with the use of a drug.
- Procuring the drug and using it takes precedence over other activities

- **Drug habituation (Psychological dependence)**

It denotes less intensive involvement with the drug, so that its withdrawal produces only mild discomfort.

- Consumption of tea, coffee, tobacco, social drinking are regarded habituating, physical dependence is absent

DRUG WITHDRAWAL REACTIONS

- Sudden interruption of therapy with certain other drugs results in adverse consequences,
 - ✓ mostly in the form of worsening of the clinical condition for which the drug was being used
- **Example:** Acute adrenal insufficiency may be precipitated by abrupt cessation of corticosteroid therapy.

TERATOGENICITY (Teratos- Monster)

- Drug to cause foetal abnormalities when administered to the pregnant mother.
- Drugs can affect the foetus at 3 stages-
 - (i) Fertilization and implantation**-conception to 17 days- failure of pregnancy which often goes unnoticed.
 - (ii) Organogenesis-18 to 55 days** of gestation most vulnerable period, deformities are produced.
 - (iii) Growth and development**-56 days onwards developmental and functional abnormalities can occur,
e.g. ACE inhibitors, Thalidomide, Warfarin, Barbiturates,.....

MUTAGENICITY AND CARCINOGENICITY

- Cause genetic defects and cancer respectively.
- Reactive intermediates which affect genes and may cause structural changes in the chromosomes
- Even without interacting directly with DNA, certain chemicals can promote malignant change in genetically damaged cells, resulting in carcinogenesis.
- Examples- anticancer drugs, radioisotopes, estrogens, tobacco.....

DRUG INDUCED DISEASES

- These are also called **iatrogenic (physician induced)** diseases, and are functional disturbances (disease) caused by drugs .
- Hepatitis by isoniazid and Rifampicin
- Peptic ulcer by salicylates and corticosteroids
- **Retinal damage by chloroquine**

•

What is gene therapy?

Thanks